

EUROPEAN COMMISSION

DIRECTORATE-GENERAL FOR COMMUNICATIONS NETWORKS, CONTENT AND TECHNOLOGY

Artificial Intelligence Office

Artificial Intelligence for Societal Good

Report on the digital transformation of communities

CNECT/2022/OP/0077 – Local Digital Twins toolbox procurement – Scale the deployment of an enabling digital infrastructure

This study has been prepared as part of the Local Digital Twins Toolbox project. The project is an initiative of the European Commission. Directorate-General for Communications Networks, Content and Technology is responsible for contract management of the Local Digital Twins Toolbox project.
For more information about this paper, please contact the following.

European Commission
Directorate-General for Communications Networks, Content and Technology
Unit A5 Artificial Intelligence for Societal Good

Authors:
Ana Correia (Deloitte), Mariana Sousa (Deloitte), Douwe Schoondorp (Deloitte), Nicolas Geusen (Deloitte), Ruben D'Hauwers (imec)
Reviewers:
Ines Ramos (Deloitte), Eliana Gerardi (Intellera), Agustina Brizuela (Intellera), Fiorella Tamburrelli (ICLEI), Ignacio Veja (Serendipity), Hugh Gardner (Capgemini)

Last update: February 2025

DISCLAIMER
The information and views set out in this publication are those of the author(s) and do not necessarily reflect the official opinion of the Commission. The Commission does not guarantee the accuracy of the data included in this study. Neither the Commission nor any person acting on the Commission's behalf may be held responsible for the use that may be made of the information contained therein. Any final decision taken on the basis of the analysis prepared by the contractor will be the result of an independent assessment by the Commission services.

Brussels: Directorate-General for Communications Networks, Content and Technology, 2025
© European Union, 2025



The reuse policy of European Commission documents is implemented by Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under a Creative Commons Attribution 4.0 International (CC-BY 4.0) licence (https://creativecommons.org/licences/by/4.0/). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

ISBN 978-92-68-33668-7 doi: 10.2759/7340057 KK-01-25-147-EN-N

Table of Contents

1. About this report 5

1.1. Purpose of this report 6

1.2. The Online Procurement Helpdesk for Smart Communities 6

1.3. Brief methodological overview 7

1.4. Key results: Overall Digital and Technical Maturity of EU Communities 8

1.5. Strategy and Organisation 12

1.6. Data Availability and Management 17

1.7. IT Infrastructure Capabilities 21

1.8. Local Digital Platform 29

2. SWOT Analysis 31

2.1. Rationale guiding the SWOT analysis 31

2.2. Strengths and Weaknesses 31

2.3. Opportunities and Threats 34

3. Policy implications and next steps for the EU Strategy towards Local Digital Twins 38

3.1. 10 Recommendations on the way forward 38

4. Annexes 41

4.1. Methodology for Digital and Technical Maturity 41

Table of Tables

Table 1 - Geographical representation of communities 41
Table 2 - Size of included communities 42
Table 3 - LORDIMAS score classification matrix 42
Table 4 - Technical maturity levels 43
Table 5 - Technical capability and sub-capability scoring 43
Table 6 - Digital Strategy maturity levels 44
Table 7 - Strategy Vision maturity levels 44
Table 8 - Digital Strategy Maturity Scoring 44

Table of Figures

Figure 1 - Number of EU communities served to the moment, per service phase 5
Figure 2 - The Online Procurement Helpdesk for Smart Communities (selected previews) 6
Figure 3 - The Helpdesk Journey of Communities 7
Figure 4 - LORDIMAS capabilities and sub-capabilities 8
Figure 5 - Number of EU communities per Digital maturity level 9
Figure 6 - Maturity scores per topic 9
Figure 7 - Technical capabilities and sub-capabilities 10
Figure 8 - Number of EU communities per Technical maturity level 11
Figure 9 - Maturity scores per topic 11
Figure 10 - Digital strategy (or equivalent) in place 13
Figure 11 - The main sectors in which EU communities are currently investing in their digitalisation 13
Figure 12 - Inclusion of a plan in the Digital strategy for the implementation of a Local Digital Platform/ Local Digital Twins 14
Figure 13 - Multidisciplinary team in place to guide the smart city journey and local digital platform/digital twin implementation 15
Figure 14 - Main types of shared digital resources among communities 16
Figure 15 - Investment priorities per sector (current and planned) 17
Figure 16 - Data Governance Framework 18
Figure 17 - Entity responsible for managing the data dimension 18
Figure 18 - Data Capabilities Scores Distribution 19
Figure 19 - Data collection sources 19
Figure 20 - Data Sources used 20
Figure 21 - Usage of tools to support GDPR compliance 20
Figure 22 - IoT Devices 23
Figure 23 - Current use of IoT Devices 24
Figure 24 – Communities' plans for implementing IoT Devices 25
Figure 25 - Responsibility of the Storage capability 26
Figure 26 - Storage solutions deployed by communities 26
Figure 27 - Responsibility of the Compute capability 27
Figure 28 - Compute solutions deployed by communities 27
Figure 29 - Disastery Recovery Plan 29

Figure 30 - Understanding the concept of LDP and LDT 29
Figure 31 - Existence of a Local Digital Platform 30
Figure 32 - Plan for a Local Digital Platform 30
Figure 33 - Rationale for the SWOT Analysis 31

1. About this report

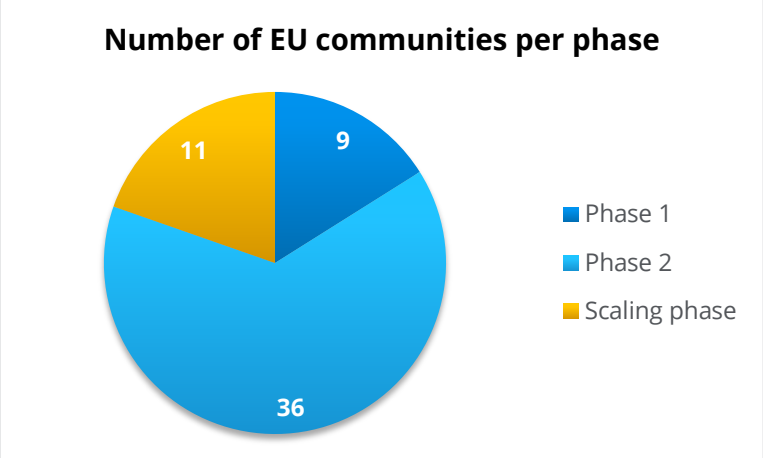
The **Digital Europe Programme** (DEP) seeks to integrate digital technologies into businesses, public administrations, and the daily lives of citizens across the European Union (EU). As part of this overarching mission, the European Commission (EC), launched support services aimed at scaling deployment of digital infrastructure and digital strategies, targeting cities and communities at the early stages of their digital transformation.

The **Online Procurement Helpdesk Facility** for smart communities was established in January 2024, with an incremental and iterative, phased approach. The project has already provided tailored support to **56 EU cities and communities across three phases**[1] of the project. The support aims to bridge the gap between current digital infrastructure capabilities and the strategic aspirations of smart cities. By conducting comprehensive assessments, the helpdesk identifies existing capacities and gaps, facilitating the creation of digitalisation roadmaps. These roadmaps guide communities in developing **local digital platforms** (LDPs) **and local digital twins** (LDTs). An LDP serves as a centralised platform for collecting data from various systems, applications, Internet of Things (IoT) devices and other data sources, while an LDT is a virtual representation of a community's physical assets, using data to help simulation models that can be updated and changed (real-time) as their physical equivalents change.

The Helpdesk includes a dedicated **Procurement Support Centre**, offering participating communities access to a repository of procurement templates and guidelines. These resources are designed to support the acquisition and deployment of essential digital tools and technologies outlined in the roadmaps as existing gaps or improvements needed in their digital infrastructure.

Recognising the diverse maturity levels of EU cities and communities, the project targeted those at the early stages of digital transition while ensuring a balanced geographical representation. By focusing on these less advanced communities, the project addresses the foundational aspects of digital transformation, laying the groundwork for scaling the enabling digital infrastructure of communities.

Figure 1 - Number of EU communities served to the moment, per service phase



Source: Consortium (Helpdesk assessment results))

[1] **Phase 1**: The focus was on offering dedicated support to a select group of 9 pilot cities. As the phase progressed, improvements were gradually made (from January 2024 to March 2024).
Phase 2: Build on the insights from phase 1 to refine the survey questionnaires and enhance the user experience of the Online Procurement Helpdesk (from May 2024 to October 2024).
Scaling phase: Use the lessons learned from phase 2 to further improve the assessments and optimise the user experience of the Online Procurement Helpdesk (from November 2024 onwards).

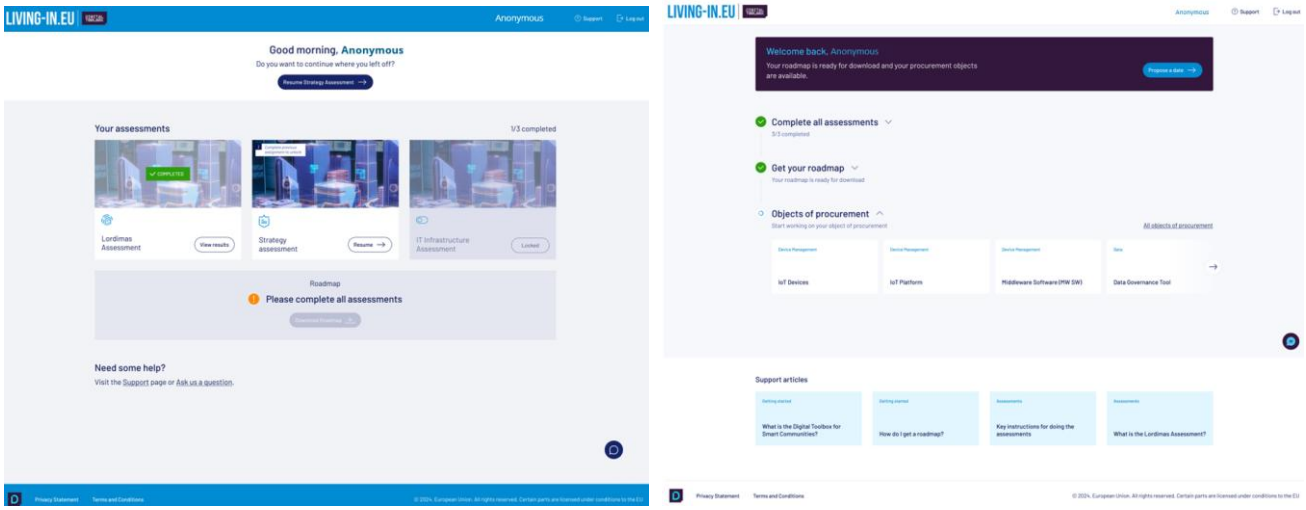
1.1. Purpose of this report

The purpose of this report is to present and analyse key findings on the current level of digital transformation across communities in the EU, based on a sample of 56 participating communities from 20 EU Member States (Annex 4.1). The document first presents the key results regarding the digital and technical maturity of EU communities. It then provides a SWOT analysis (Strengths, Weaknesses, Opportunities, and Threats) based on these results. Finally, the report outlines potential policy implications and next steps for the EU strategy to promote the wide-scale adoption of LDPs and LDTs.

1.2. The Online Procurement Helpdesk for Smart Communities

This section introduces the Online Procurement Helpdesk service. The Helpdesk provides free-of-charge consulting services to EU communities in their digital transformation journey, offering guidance in three critical dimensions, namely in assessing communities' current digital and technical maturity, providing a tailored digitalisation roadmap, and offering procurement support for the deployment stage. It promotes the alignment with the EU Digital Decade's goals, and places interoperability at the centre of this transformation, by considering the use of open standards and Minimal Interoperability Mechanisms (MIMs) when selecting and implementing certain smart city solutions and technologies. This approach ensures that communities across Europe can adopt LDTs faster and with their gradual evolution in mind, ensuring sustainable investments and avoiding vendor lock-in situations.

Figure 2 - The Online Procurement Helpdesk for Smart Communities (selected previews)

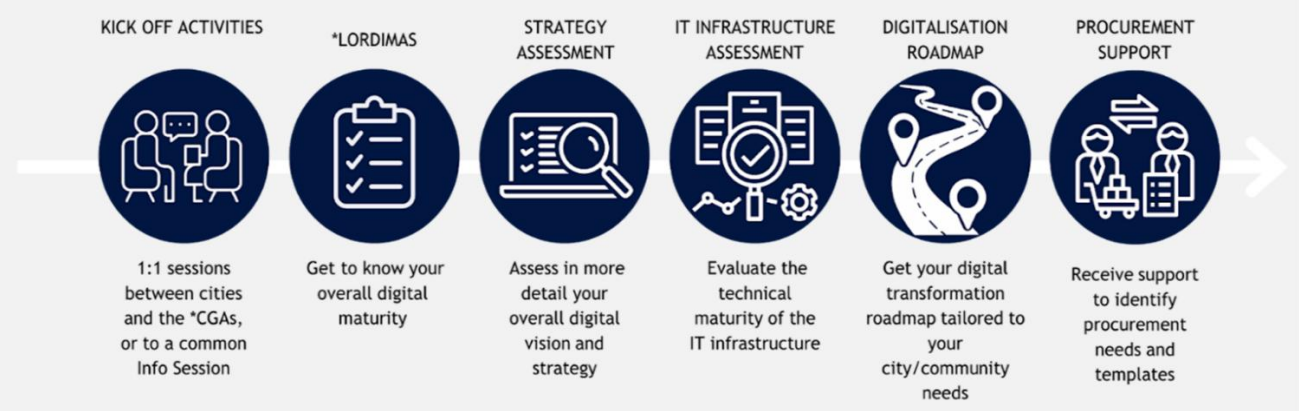


Source: [LDT Toolbx]

The Helpdesk Journey of EU Communities

The primary objective of the journey is to equip communities with the tools and methodologies needed to assess their "as-is" situation and conclude with the roadmap of initiatives needed to progress their digital infrastructure towards an LDP and LDT, including procurement support for the deployment stage afterwards. This structured journey includes six key organisational and analytical steps:

Figure 3 - The Helpdesk Journey of Communities



Source: Consortium

- **Kick-Off Activities**: Following the submission of an Expression of Interest (EoI), a session is organised by the city guiding agent (CGA) with city representatives to present the project and related services.
- **LORDIMAS**[2] (Local and Regional Digital Maturity Assessment): Once provided access to the Online Helpdesk, city officials begin by completing the LORDIMAS assessment to evaluate their overall digital maturity to get an initial "snapshot" of their digital maturity.
- **Strategy Assessment**: Next, city officials conduct an assessment to align their strategic priorities and future ambitions with the capabilities of LDPs and LDTs.
- **Information Technology (IT) Infrastructure Assessment**: City officials then conduct a self-assessment of their core IT capabilities to determine technical readiness for developing LDPs and LDTs.
- **Digital Transformation Roadmap**: Based on the results from the previous assessments, communities receive a tailored digital transformation roadmap detailing phases, initiatives, and milestones for their digital journey towards an LDT.
- **Procurement support**: Communities receive support (procurement templates and guidelines) based on current digital infrastructure gaps.

1.3. Brief methodological overview

The methodologies used for the analysis include LORDIMAS as an overarching framework for measuring digital maturity and a more detailed IT infrastructure assessment for a deeper investigation into the core IT capabilities that are key to the development of an LDP and LDT. A Strategy assessment ensures the alignment of strategic and IT priorities. These methodologies are described in **Annex 4.1**, including more details on the sample of 56 participating communities up until January 2025.

[2] LORDIMAS is developed by ESPON in cooperation with the European Committee of the Regions, European Commission, Living-in.EU supporters and signatories and the Go Li.EU consortium, that enables digital maturity self-assessment of communities.

1.4. Key results: Overall Digital and Technical Maturity of EU Communities

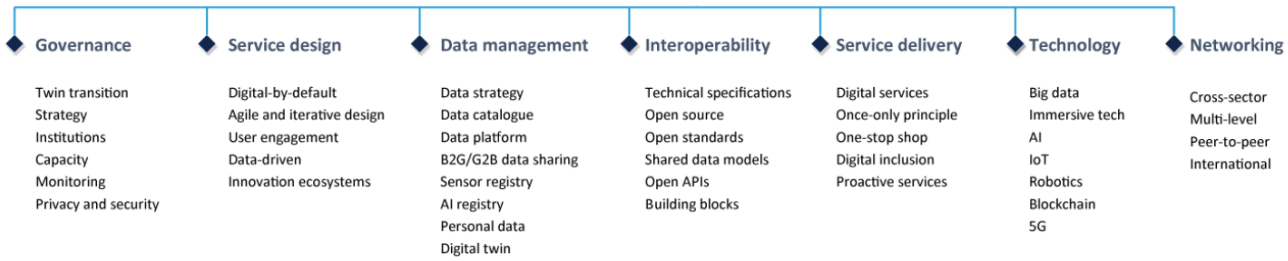
In this subsection, we start by providing the overall digital maturity scores of the 56 communities covered by this analysis, based on the LORDIMAS assessment. We then zoom in on the technical maturity results to have a more detailed picture of the strengths and weaknesses of communities' digital infrastructure.

It is important to note that, as the target audience of this project is mostly cities and communities at their initial stages of digital transformation, it is expected that the scores reflect that "self-selection" effect.

1.4.1. Digital maturity of EU communities

The digital maturity of EU communities reflects their ability to adapt, integrate, and optimise critical digital capabilities necessary for digital transition. LORDIMAS assesses seven dimensions, namely Governance, Service design, Data management, Interoperability, Service delivery, Technology, and Networking, each comprising a set of sub-capabilities.

Figure 4 - LORDIMAS capabilities and sub-capabilities



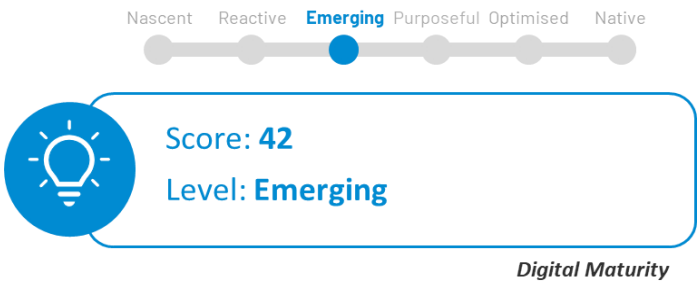
Source: LORDIMAS

Following the assessments, LORDIMAS will assign scores to each capability and sub-capability using a defined methodology (see Annex 4.1).

Overall digital maturity score

The overall digital maturity score of EU communities is derived by calculating the average of all capability scores. This score provides a comprehensive measure of EU communities' progress in digital transformation across the seven evaluated capabilities.

*Key message #1: Overall digital maturity marked by an **emerging** digital transition*

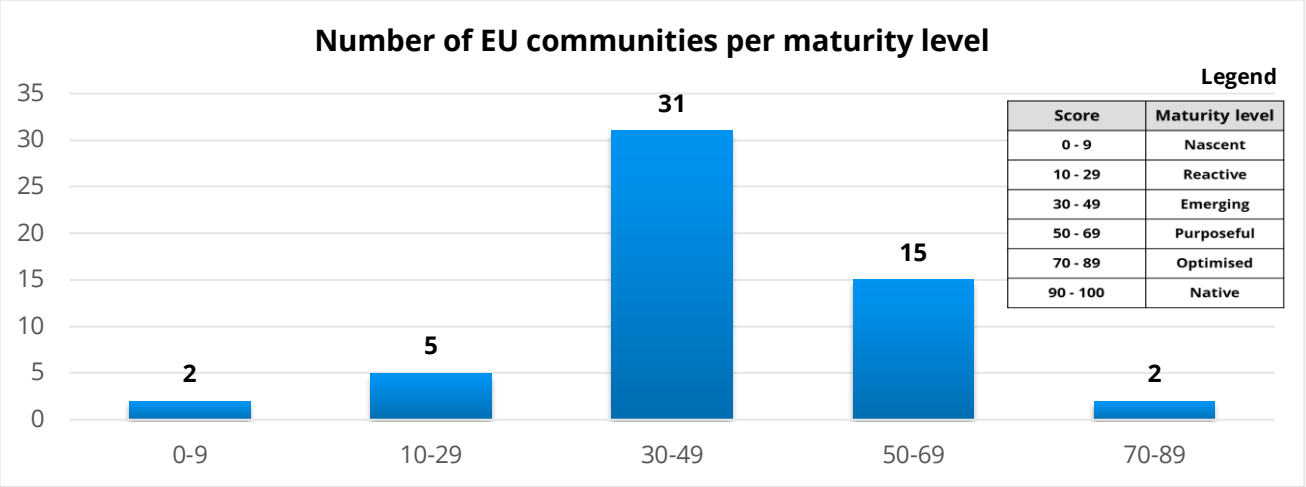


The current **overall digital maturity score of our sample of communities stands at 42, placing EU communities at the *Emerging* level**, using LORDIMAS terminology. This classification signifies the early stages of digital transition[3], indicating that while there are ongoing efforts toward digitalisation, substantial improvements are still needed to

[3] LORDIMAS. The 6 stages of Digital Maturity. LORDIMAS_EN.

achieve higher levels of maturity. This score confirms that low-medium mature communities were, in fact, those mostly targeted by this project.

Figure 5 - Number of EU communities per Digital maturity level



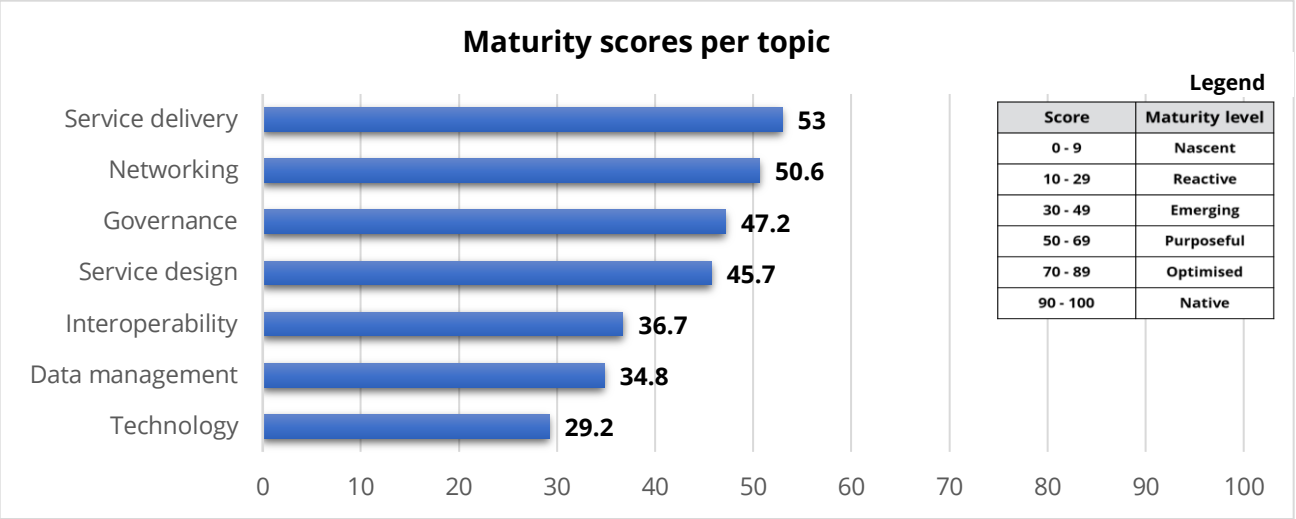
Source: LORDIMAS

56% of EU communities fall within the Emerging level, with scores ranging from 30 to 49. 27% of communities are at the Purposeful level, scoring between 50 and 69, while 9% have scores between 10 and 29. This is in line with the expected results, as most cities targeted in the scope of this initiative will be situated at the beginning of their digital transition.

Maturity scores per topic

Key message #2: Service delivery and Networking have the highest maturity scores

Figure 6 - Maturity scores per topic



Source: LORDIMAS

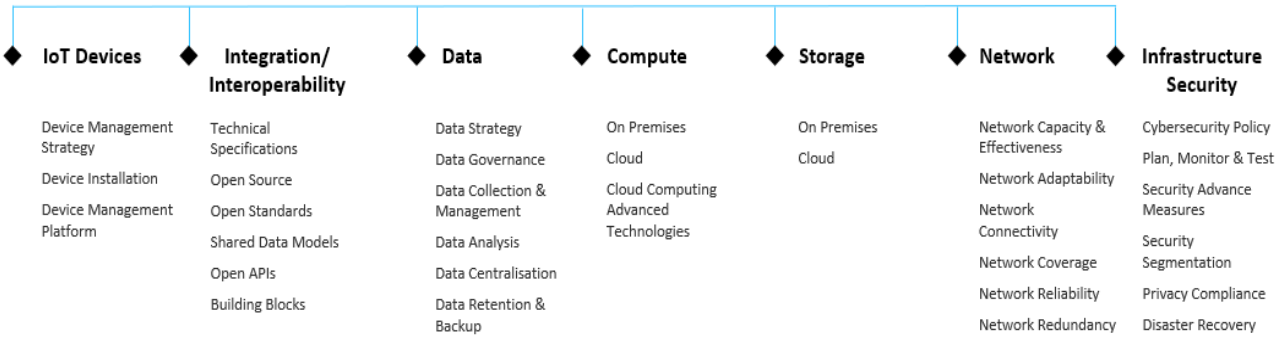
The assessment of maturity scores across digital capabilities provides valuable insights into the digital transition within EU communities. **Service delivery and Networking, the capabilities with the highest scores, are classified as Purposeful**, reflecting good efforts to implement practical and tangible digital

solutions for end-users and some level of engagement with other cities and initiatives as well. Governance, Service design, Interoperability, and Data management are categorised as Emerging, indicating low but progressing digital transition. On the other hand, at the lower extreme is the Technology dimension (29.1), with **slow progress in terms of the adoption or uptake of technologies** to support the operations of the community. In particular, the score is influenced by low Robotics and Blockchain maturity scores, with 13.3 and 11.5, respectively, while 5G, Big Data and Artificial Intelligence (AI) registered scores of 31.1, 32.6 and 41, respectively. Overall, this suggests that greater focus should be placed on investing in digital technologies as they are essential for the development of LDP and LDT. The lack of adoption is typically explained by budget constraints, lack of skilled staff and/or unawareness of the costs and benefits of such technologies.

1.4.2. Technical maturity of EU communities

The overall technical maturity of EU communities is based on insights from the LORDIMAS, the Strategy assessment (analysed in detail in the next subsection), and the IT Infrastructure assessment, which evaluates seven core IT capabilities essential to develop an LDP and LDT. The capabilities encompass IoT Devices, Integration/ Interoperability, Data, Compute, Storage, Network, and Infrastructure Security, each comprising a set of sub-capabilities.

Figure 7 - Technical capabilities and sub-capabilities



Source: Consortium

Overall technical maturity score

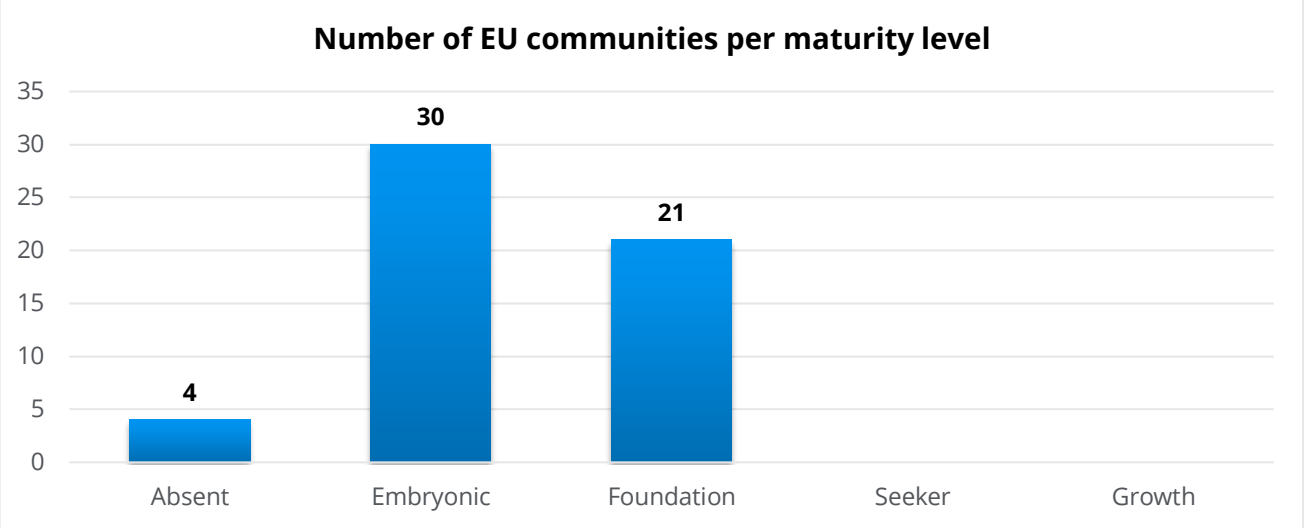
The overall technical maturity score of EU communities is derived by calculating the average of all capability scores.

*Key message #3: Overall technical maturity marked by **foundational** technical transition*



The current **overall technical maturity score is 3, positioning the EU communities at the Foundation level**. This classification indicates that, on average, communities have established basic technological capabilities. The overall technical maturity classification differs from the capabilities classification, which is detailed in Annex 4.1.

Figure 8 - Number of EU communities per Technical maturity level



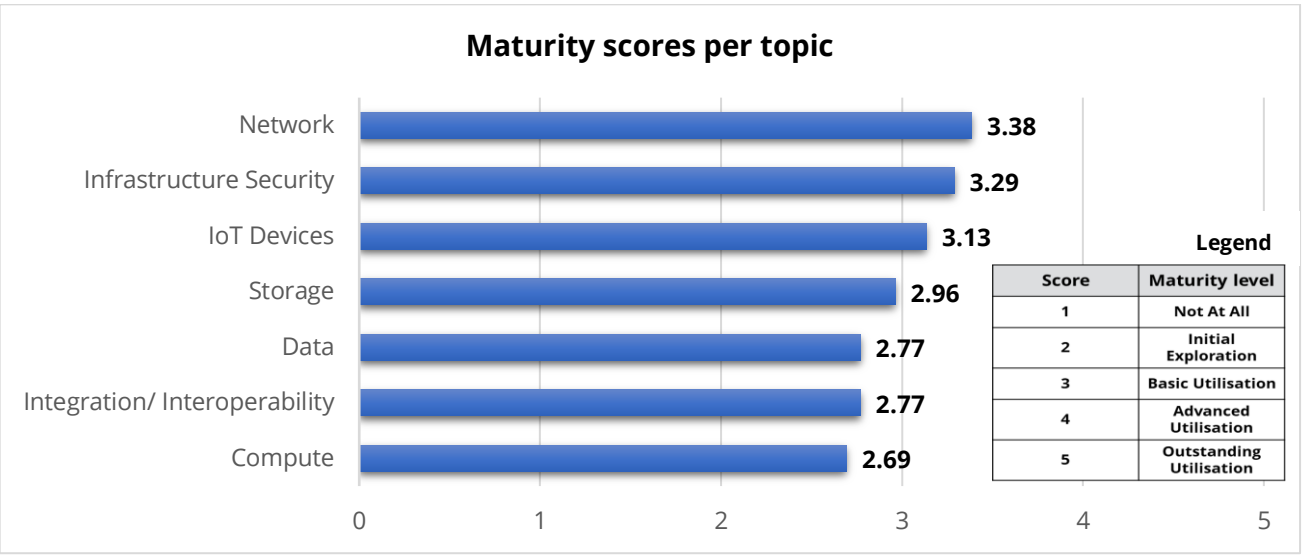
Source: Consortium (Helpdesk assessment results)

54% of EU communities fall within the Embryonic level, making it the most prevalent maturity level. 38% of communities are at the Foundation level, while 7% are at the Absent level, indicating poor technological capabilities. However, these communities demonstrate a willingness and/or political commitment to embark on their digital transformation journey, including adopting smart solutions. The results are in line with the expected outcome, reflecting the early stages of the communities that have been part of this initiative.

Maturity scores per topic

Key message #4: Network has the highest maturity score

Figure 9 - Maturity scores per topic



Source: Consortium (Helpdesk assessment results)

The analysis provides valuable insights into the state of technical maturity across key capabilities within the assessed EU communities. **All capabilities fall within the range of Basic Utilisation**, as reflected in

the overall technical maturity score, which shows that communities have a functional understanding and application of these capabilities to fundamental tasks.

However, **Network** (e.g., coverage, capacity, adaptability, redundancy, 5G readiness, speed) **stands out with stronger performance**, approaching Advanced Utilisation. This suggests that communities tend to demonstrate a deeper understanding of the complexities of this capability and can leverage Network features to further scale their infrastructure in the future.

In contrast, **Compute** (e.g., computational resources, compute management, provisioning, cleanup, policies/procedures) **has the lowest score**, with Integration/Interoperability and Data closely following. The Compute scores highlight the need for greater efforts to enhance communities' systems and network capabilities for processing and handling data, as well as running applications more effectively. Additionally, **Integration/Interoperability and Data should be considered critical** to help EU cities and communities overcome challenges such as working in data silos and inefficiencies in system and application integration and communication— as both are fundamental enablers of digital services and smart city development.

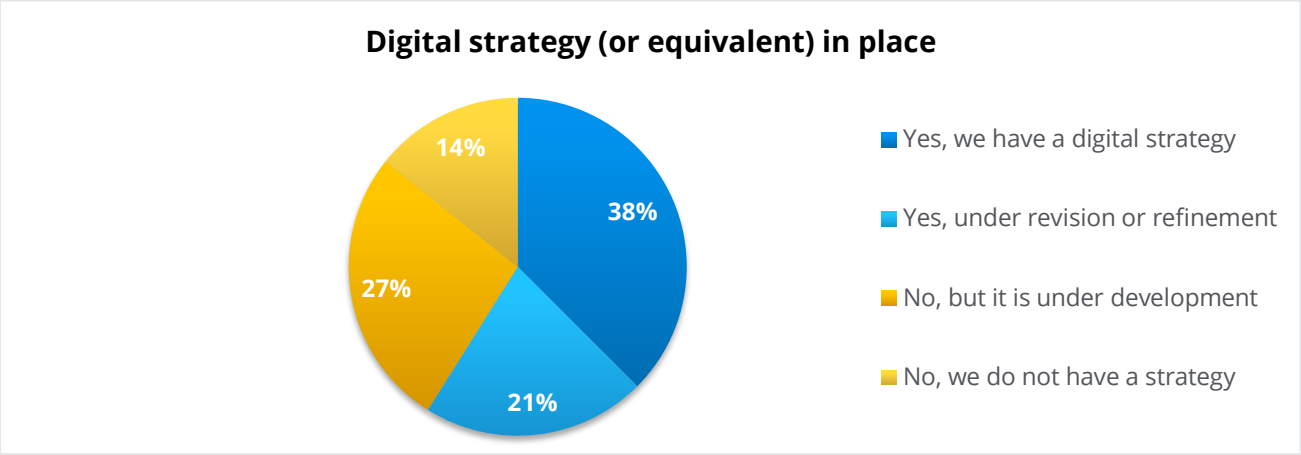
1.5. Strategy and Organisation

While the core of this service focuses on assessing a community's current technical infrastructure, strategic priorities play a critical role in shaping their digital transformation journey. Recognising this, the analysis extends beyond technical capabilities to include strategic and organisational aspects, ensuring a holistic understanding of each city's digital maturity.

By gathering insights on governance structures and digital strategies, the following section provides a comprehensive picture of how communities approach digital transformation. The ultimate objective is to understand where EU communities stand, relating both their current ("as-is") and desired ("target") digital state, aligning with their strategic objectives.

1.5.1. Adoption of a digital strategy across EU communities

The first key step in a community's digital transformation journey is the creation of a Digital Strategy. Based on the findings from an assessment conducted with participating communities, it has become a standard practice, with only **14%** of communities reporting that they **lack a digital strategy** altogether. Additionally, **26% of the communities** surveyed are **currently in the process of developing one**. Another relevant aspect relates to the fact that most communities have their digital strategy published (42%) or intend to publish it (23%) officially. These results underline that the need for communities to develop strategies has become increasingly clear, as municipalities must design plans that will guide the adoption of smart technologies and drive their digital transformation efforts.



Source: Consortium (Helpdesk assessment results)

In this process of strategical definition, a critical challenge was reported: achieving a unified vision across the organisation. For 45% of the participating communities that are developing or updating their strategies, **internal fragmentation remains a significant obstacle**. With departments within cities often working on separate initiatives, aligning everyone around a shared vision for digital transformation can be difficult.

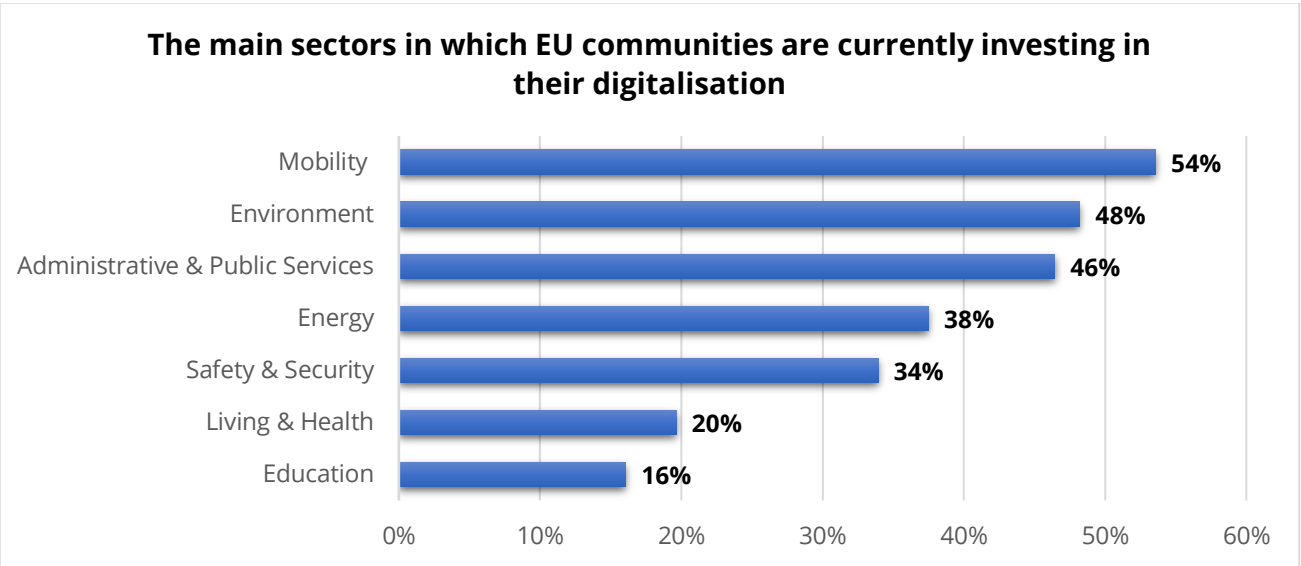
Key message #5:

45% Of the communities with a Digital Strategy, report that their **strategic objectives are dispersed** across different documents, departments or sectors.

Current Strategic Priorities for Digitalisation

Communities' digital investment efforts are currently concentrated in three key sectors: **Mobility (54%)**, **Environment (48%)** and **Administrative & Public Services (46%)**, followed by Energy, and Safety & Security. These areas reflect municipalities' current focus on enhancing urban mobility and sustainability.

Figure 11 - The main sectors in which EU communities are currently investing in their digitalisation



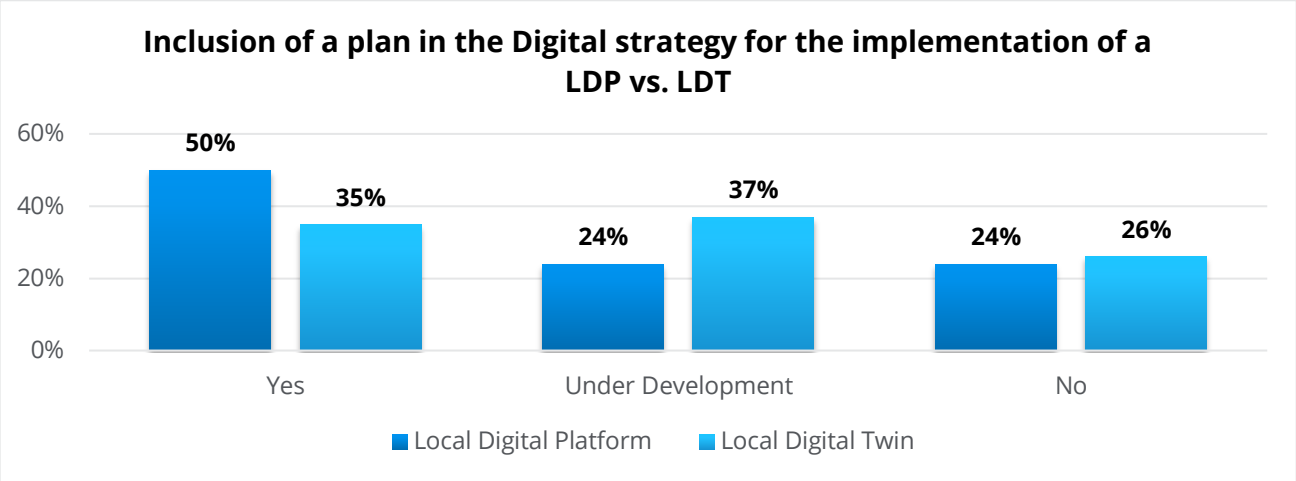
Source: Consortium (Helpdesk assessment results)

1.5.2. Local digital platforms and local digital twins as emerging priorities

As cities and communities develop their digital strategies, two key priorities are emerging from a digital infrastructure standpoint, namely, the development of LDPs and LDTs. Our analysis indicates that **50% of participating communities have already integrated LDP development as a priority into their digital strategies**, with an additional **24% planning to include them in the future**. LDPs are essential for communities to deliver digital services to their residents and businesses, and to generate an integrated view supporting the operations in cities, creating an interconnected digital environment.

LDTs have not yet been widely referenced in digital strategies. **Only 35% of the communities currently feature LDTs in their digital strategies**. However, an additional 37% are actively working toward their integration, indicating growing momentum for their adoption. While LDTs are not yet a standard feature, this trend suggests a shift in priorities and a potential future focus on LDTs as part of digital transformation efforts.

Figure 12 - Inclusion of a plan in the Digital strategy for the implementation of a Local Digital Platform/ Local Digital Twins



Source: Consortium (Helpdesk assessment results). All percentage values are rounded to one decimal place.

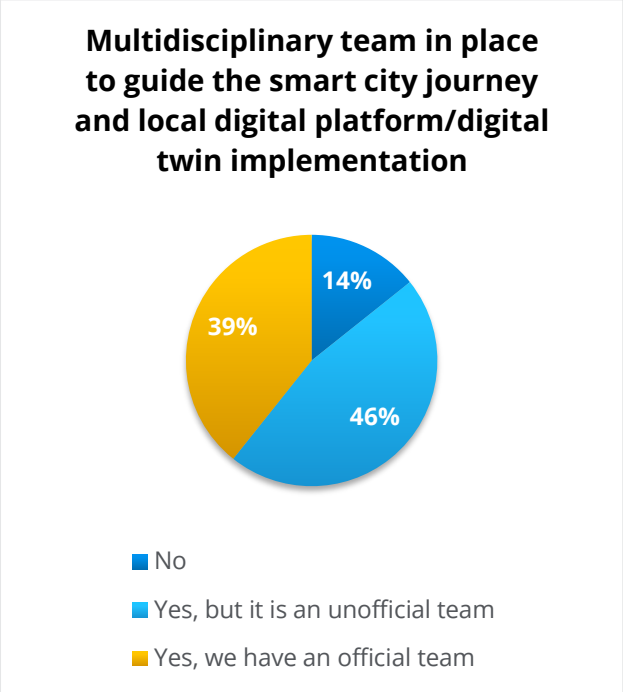
This trend underscores the critical timing for communities to develop a strategic approach to LDTs to future-proof their digital infrastructure and to strengthen the use of LDTs to support operational day-to-day decisions and more informed policy decisions.

1.5.3. Organisational structures powering the digital transformation

For a digital strategy to succeed, clear governance is essential. According to the survey, **68% of communities have already established governance structures to oversee their digital strategies**. The roles of Chief Technology Officers (CTOs) and Chief Digital Officers (CDOs) are becoming prominent figures in city administrations, ensuring high-level leadership and oversight of digital projects. These leadership roles are critical for guiding digital transformation efforts across the city, especially when they impact various city domains.

Additionally, when asked concretely about their smart city journey and LDTs adoption, 86% of the communities have reported having established dedicated multidisciplinary teams, such as a **Smart City Office or Digital Transformation Office**, to guide their digital transformation journey. **However, more than half (54%) of these communities have not yet officially formalised these teams** within their governance structure, indicating a gap in establishing formal authority and responsibility.

Figure 13 - Multidisciplinary team in place to guide the smart city journey and local digital platform/digital twin implementation



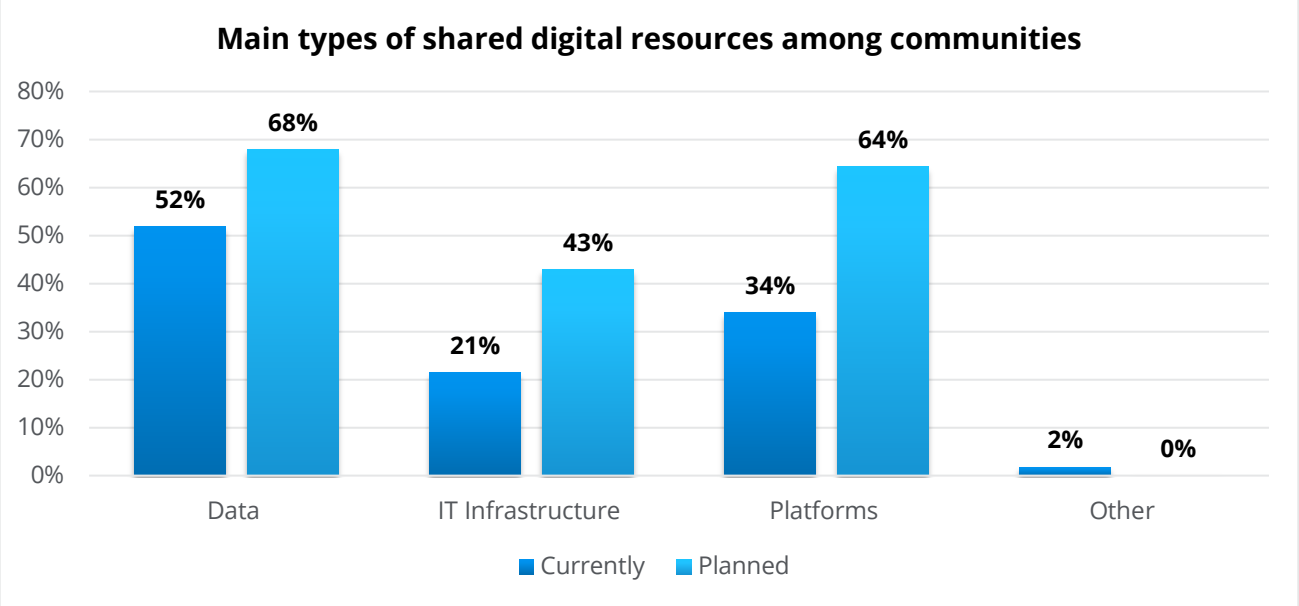
Source: Consortium (Helpdesk assessment results)

Interest in collaborating is also visible, as **82% of communities plan to share digital resources** with other cities within their region or metropolitan area, marking a significant increase from the **56% that currently have shared resources in place**. This shift may indicate a commitment to fostering **inter-community cooperation** and optimising digital infrastructure through shared solutions. Notably, this trend extends across various resource types, including **data and digital platforms**, while a particularly strong increase is observed in the **intention to share IT infrastructure**, such as storage, computing, networks and security solutions, devices and servers, highlighting a strategic move towards creating synergies in common digital ecosystems. As many of these digital infrastructure investments tend to be costly both from an implementation and maintenance perspective, the intention to share these investments and resources, including through joint procurement or co-funding, is particularly relevant to smaller municipalities that often struggle to have sufficient financial and human resources to invest and lead these investments on their own.

Key message #6:

82% of communities plan to **share digital resources** with others to support IT infrastructure development.

Figure 14 - Main types of shared digital resources among communities



Source: Consortium (Helpdesk assessment results). All percentage values are rounded to one decimal place.

1.5.4. Future outlook: performance tracking and emerging strategic priorities

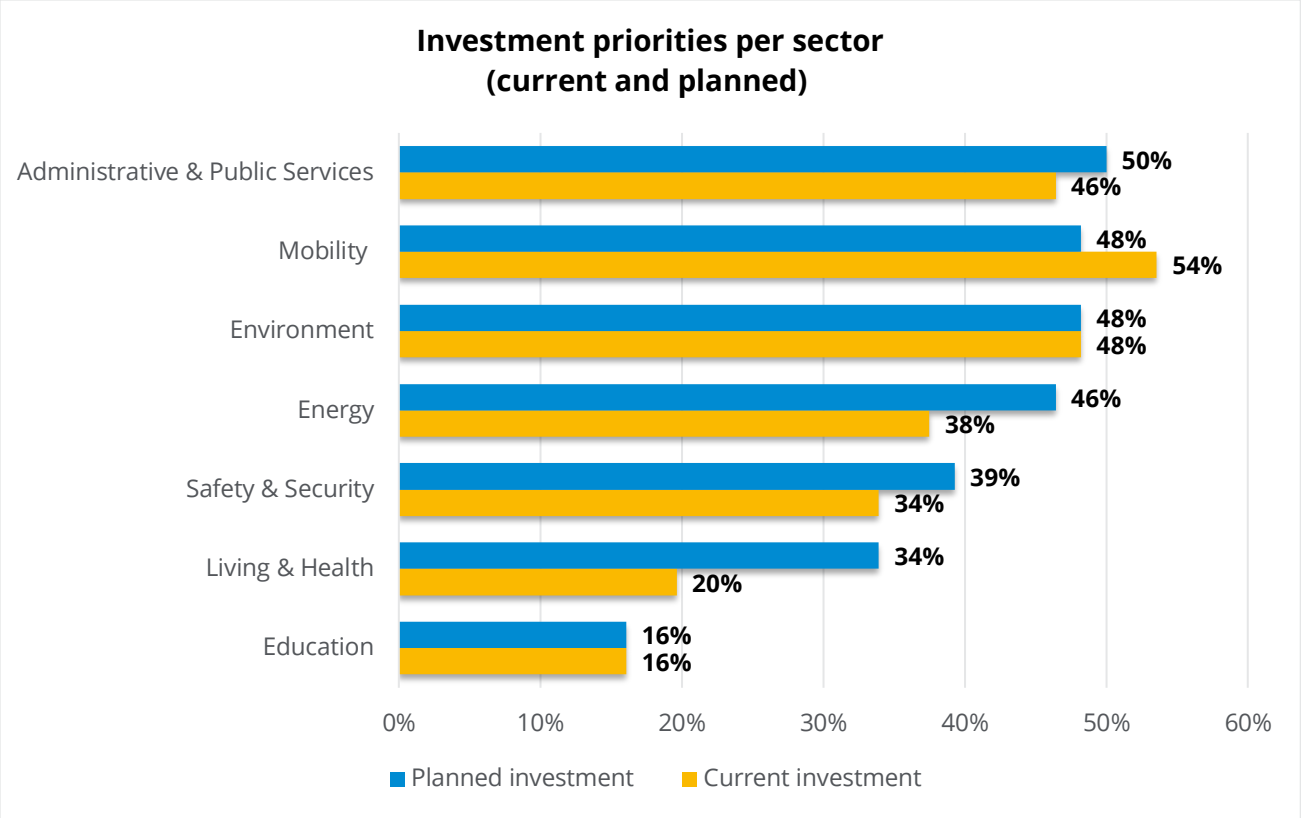
In the assessment conducted, it became evident that **monitoring strategic ambitions and digital initiatives** remains an area in which EU communities have **significant room for improvement**. **41% of communities in our sample reported that they are actively working on defining and establishing key performance indicators (KPIs)** to track their digital performance. However, among the 23% of communities that have already defined KPIs, transparency and reporting remains a key challenge, as **55% still lack a dedicated report and monitoring mechanism**, such as a dashboard to present and communicate these performance indicators effectively. This highlights the need for enhanced tracking practices that not only measure progress but also ensure visibility and accountability in digital transformation efforts.

Future Investment Priorities

When looking ahead, communities indicate a smooth shift in digital investment priorities. Sectors expected to receive increased focus include **Living & Health** (a 14-percentage point increase) and **Energy** (a 9-percentage point increase), followed by positive growth in **Administrative & Public Services** and **Safety & Security**. This evolving landscape highlights a growing commitment to enhancing citizens' quality of life, advancing sustainable energy solutions, and further modernising public services.

The bar chart below illustrates the evolution of digital investment priorities among EU communities, comparing the sectors where communities are currently allocating resources versus those identified for future investment. The percentages reflect the proportion of communities that selected each sector as a focus area.

Figure 15 - Investment priorities per sector (current and planned)



Source: Consortium (Helpdesk assessment results)

1.6. Data Availability and Management

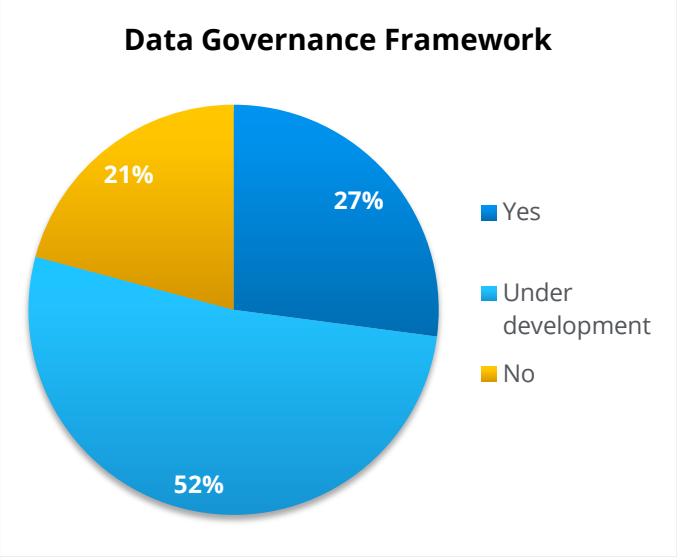
Effective data management is crucial for optimising community services and enhancing decision-making. This section assesses how EU communities collect, store, organise, and share data across service sectors, ensuring accessibility for connected decision-making. It evaluates key data capabilities and governance processes to determine how well communities utilise data in their operations. This section evaluates the interoperability of community infrastructure and examines personal data management with a focus on privacy and regulatory compliance. To understand data maturity, we examined communities' progress in data strategy, governance, collection, management, and analysis.

1.6.1. Data strategy and governance

Overall, only **31% of the surveyed EU communities have an active data strategy** that coordinates the management, sharing, interoperability and (re)use of data. However, over half of the communities are in the stages of consideration or development of a data strategy, which shows most communities understand the importance of a data strategy.

Similarly, a limited number of communities have established a data governance framework to guide data management across departments and citywide operations. Currently, **27% have implemented such a framework**, while 52% are in the development phase. However, 21% of communities lack a data governance framework altogether. This suggests that while many communities acknowledge its importance, they may need assistance in its implementation.

Figure 16 - Data Governance Framework

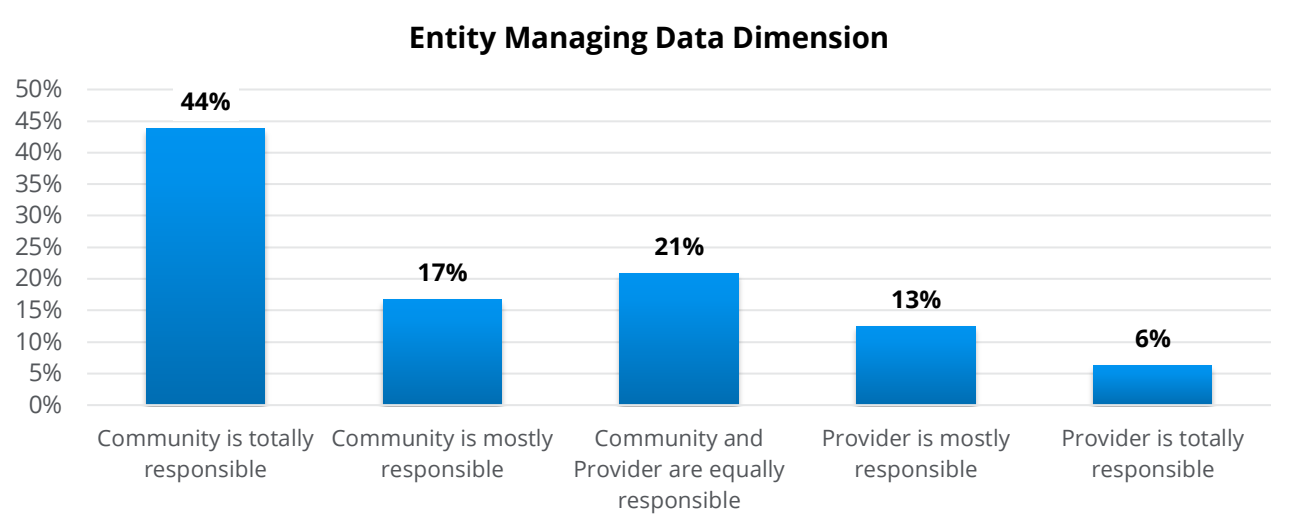


Source: Consortium (Helpdesk assessment results)

To support the implementation, management, and enforcement of data governance processes, some communities leverage tools like data catalogues, Master Data Management (MDM) tools, and data quality solutions. Currently, 46% of communities have adopted such tools, while the majority (54%) have not. Specifically, only 31% of communities have a data catalogue.

Among the communities surveyed, 44% assumed responsibility for managing the data dimension (mostly) themselves. In 19% of the communities, an external provider is either fully or mostly responsible. Additionally, in 21% of the communities, responsibility is shared equally between the community and the provider. **This indicates that 61% of communities play a significant role in managing data**, whereas only 19% outsource this responsibility.

Figure 17 - Entity responsible for managing the data dimension



Source: Consortium (Helpdesk assessment results). All percentage values are rounded to one decimal place.

Key message #7:

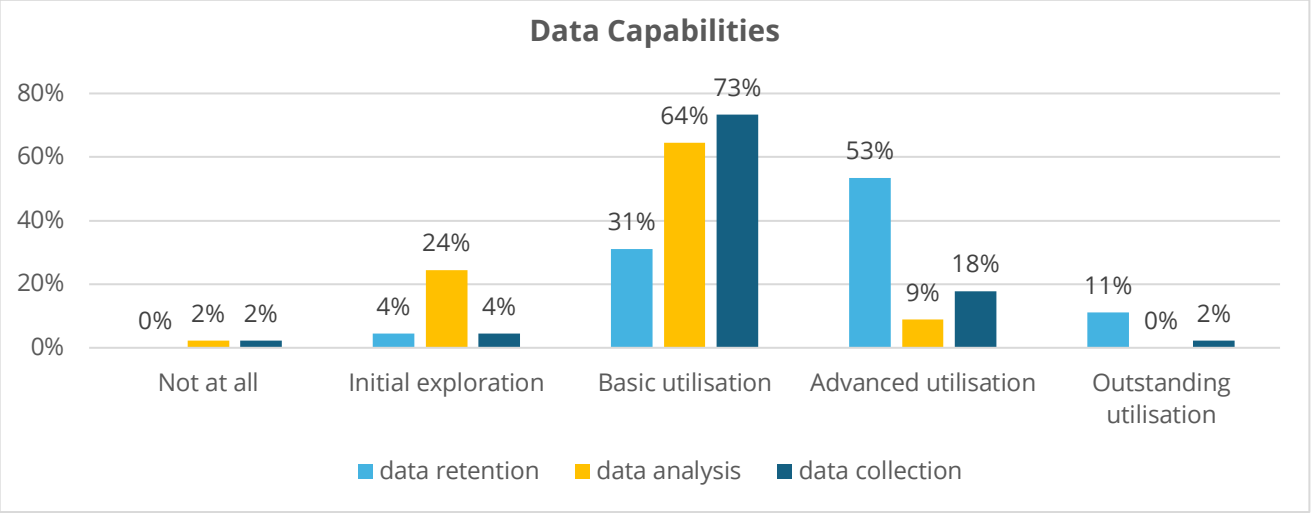
61% Of communities a play a crucial role in managing their own data, whereas only 19% outsource this responsibility.

1.6.2. Data collection, management and retention capabilities

Data collection and management capabilities, data analysis, and data retention were benchmarked to assess their maturity across communities. Among these, **data collection and management, and especially data retention, emerged as the most developed capabilities**, indicating that many communities have established processes for gathering and storing data effectively. However, the data

analysis capabilities of some communities lag behind, suggesting that while communities are successfully collecting and retaining data, they face challenges in extracting insights. This gap highlights the need for improved analytical tools, skills, and strategies to maximise the value of collected data.

Figure 18 - Data Capabilities Scores Distribution

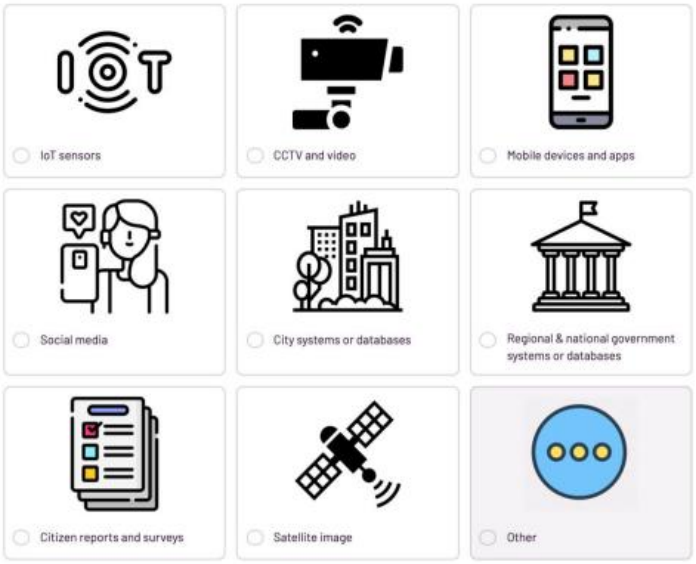


Source: Consortium (Helpdesk assessment results). All percentage values are rounded to one decimal place.

Data analysis is the capability which is developed the least in communities. 26% of the communities are lagging in these skills, and only 9% of the communities show advanced utilisation. When examining data analysis skills at a basic utilisation level or higher, 56% of communities report using data analysis tools to process, interpret, and extract meaningful insights. However, only 29% of communities have established KPIs for data analysis. This aligns with the generally average to low capability levels in this area, indicating room for improvement.

In data collection, approximately 20% of communities show proficiency with advanced or outstanding utilisation, 73% of communities indicated basic utilisation, and only 6% of communities lag behind. Many communities utilise a diverse range of sources, with **69% of communities relying on more than five data collection sources**:

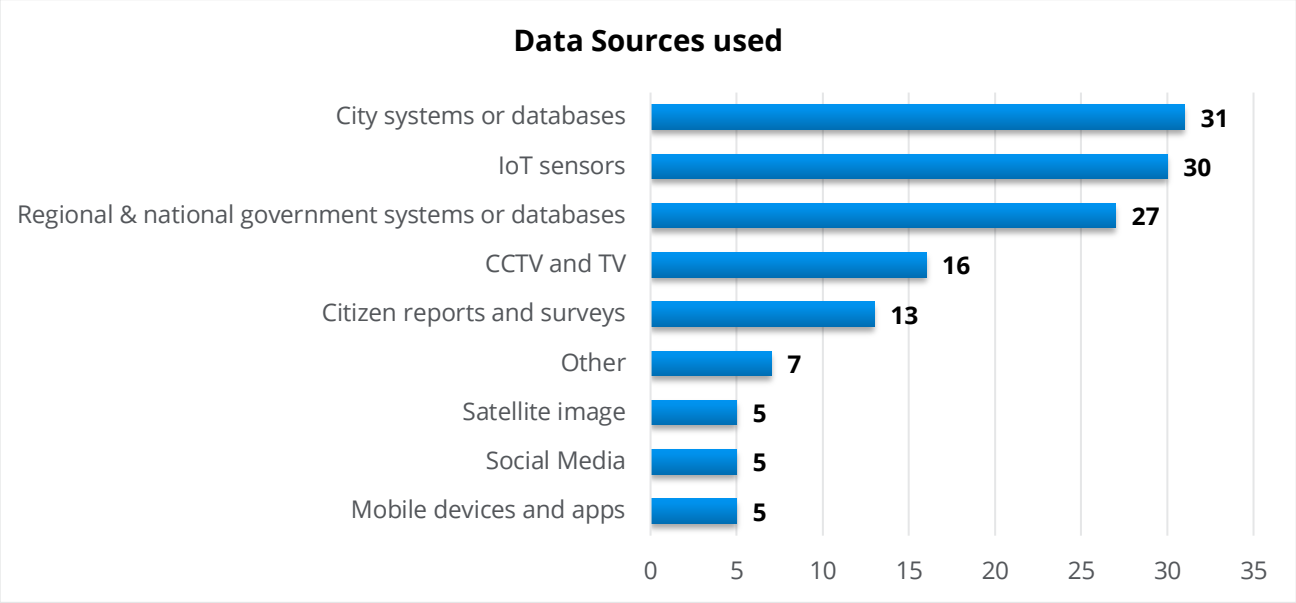
Figure 19 - Data collection sources



Source: Consortium (based on various experiences with smart city projects)

The **most common sources include community systems or databases, regional and national government systems, and IoT sensors**. Although IoT sensors are used in several communities, only 17 have a sensor registry, while the majority (75%) of communities utilise IoT sensors (see section 1.7.1). This demonstrates that communities have access to a broad array of data sources and the capability to collect information effectively. However, they may lack an overview of which sensors they have on which locations.

Figure 20 - Data Sources used



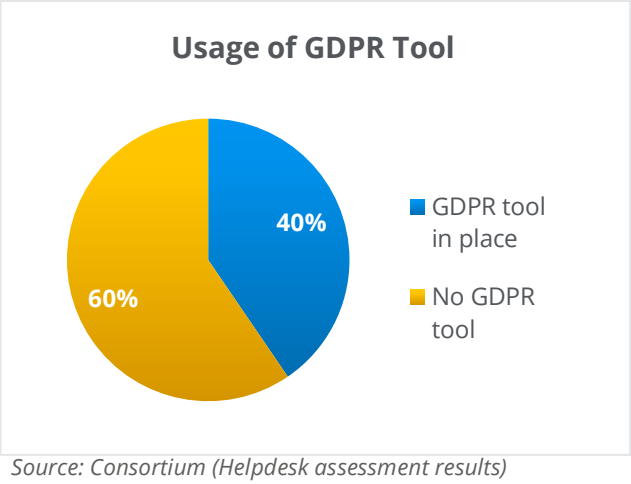
Source: Consortium (Helpdesk assessment results)

In contrast, **data retention stands out as the highest-rated capability**, with several communities reaching an advanced or outstanding utilisation level and only 4% of communities slightly lagging. 60% of communities achieved a retention coverage of 71% to 100%. Meanwhile, data centralisation varies significantly among communities: 50% of communities have a centralisation rate below 35%, while another 50% exhibit higher levels of centralisation.

1.6.3. Personal data management

Among the communities analysed, 76% have adopted privacy-enhancing technologies that enable data collection, processing, analysis, and sharing while maintaining privacy safeguards. All of those cities report to be compliant with GDPR policies and procedures, however, its application remains inconsistent. Only 40% of communities have implemented tools that support compliance with GDPR policies or procedures. Transparency remains a challenge, as only 26% of the analysed communities provide citizens with a practical way to review which personal data has been collected about them (based on LORDIMAS results). These findings highlight the need for more comprehensive data governance strategies to ensure privacy rights and

Figure 21 - Usage of tools to support GDPR compliance



Source: Consortium (Helpdesk assessment results)

transparency are upheld while leveraging digital platforms.

Key message #8:

76% Of communities have adopted **privacy-enhancing technologies**.

1.6.4. Interoperability

Interoperability is a key factor in enabling seamless data exchange and integration across urban digital systems. While several frontrunner communities have well-developed interoperability frameworks and adhere to established standards, many others are still lagging. Several communities are in the process of developing interoperability measures, reflecting a growing awareness of its importance. **37% of the communities have well-defined technical specifications to ensure interoperability**. The adoption of open standards and interoperability requirements in digital infrastructure procurement is more widespread, with 50% of communities implementing these measures and 18% of communities in the development phase.

The usage of EU digital infrastructure is an opportunity, as only 15% of communities reported having their digital infrastructure based on MIMs Plus. In contrast to MIMs, MIMs Plus incorporates key European specifications and initiatives (the "Plus"). Therefore, the promotion of standardised integration is an opportunity for these communities. Electronic Identity (eID) is the most widely implemented EU digital infrastructure component, adopted by 74% of the communities. 13% of communities use building blocks from the DEP[4] (e.g., eDelivery, eSignature, and eInvoicing), to enhance their digital infrastructure. These insights highlight both progress and gaps in urban interoperability efforts, emphasising the need for broader adoption of open standards and interoperable solutions to support sustainable investments and the gradual evolution of systems and platforms.

Key message #9:

37% Of the communities have well-defined technical specifications to ensure interoperability

1.7. IT Infrastructure Capabilities

In the modern era of digital transformation, IT infrastructure capabilities form the cornerstone of any city's digital development strategy. These capabilities encompass a wide range of technologies and systems, including IoT devices, various digital applications, and the essential components of storage, compute, and network infrastructure. IoT devices, such as sensors, cameras, and smart meters, provide real-time data from a multitude of sources, enabling cities to monitor and manage urban infrastructure, environmental conditions, and public services more effectively. Digital applications leverage this data to offer actionable insights and automated responses to dynamic urban challenges, ranging from smart traffic management and energy-efficient building controls to public safety platforms and citizen engagement tools.

The backbone of this digital ecosystem is composed of robust storage, compute, and network capabilities. Storage solutions, whether on-premise or cloud-based, ensure that vast amounts of data are securely retained and easily accessible. Compute capabilities, including on-premise servers and cloud computing, provide the necessary processing power to analyse data and run complex simulations.

[4] European Commission. Digital Building Blocks for Europe.

Network infrastructure, encompassing both wired and wireless technologies, facilitates the seamless transmission of data between devices, systems, and stakeholders. Together, these IT infrastructure capabilities enable communities to harness the power of data, driving efficiency, sustainability, and innovation, ultimately improving the quality of life for all residents.

1.7.1. IoT devices, Systems, and Applications

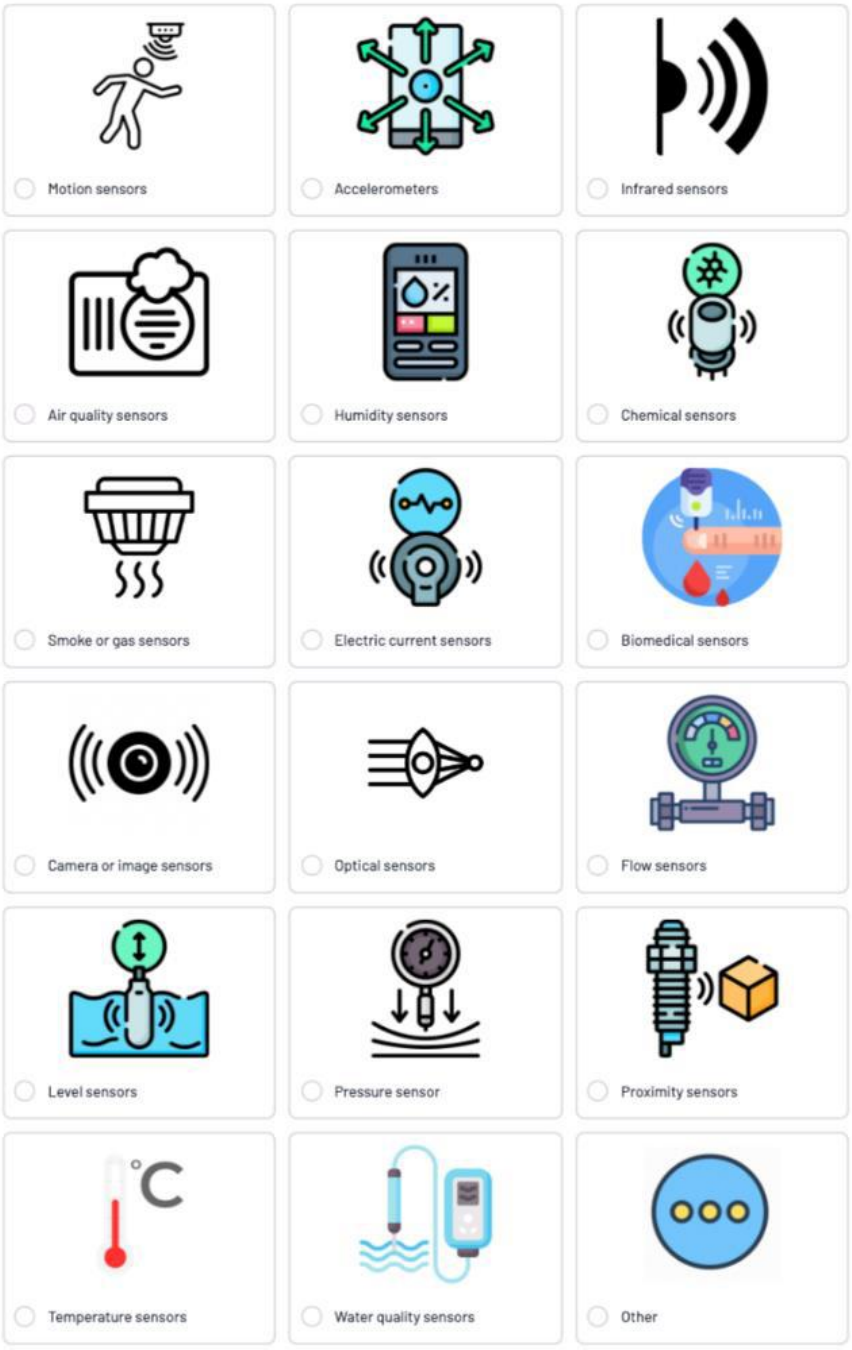
The essence of many digital advancements lies in the ability to measure or monitor a specific part of the city and to be able to represent this in a digital environment where the measurement can be used and interpreted. There are multiple ways in which this can be accomplished, but in the context of the path towards an LDT, IoT sensors and devices are often used for real-time data collection. And while data collection has some merits in itself, it is the use of this data and different systems and applications where the real value is situated.

The prevalence of different IoT devices has been asked, as well as the plans to invest and integrate additional sensors. Additionally, their use in major applications across seven sectors has been investigated. These sectors are:

- Mobility
- Energy
- Environment
- Living & Health
- Education
- Safety & Security
- Administrative & Public Services

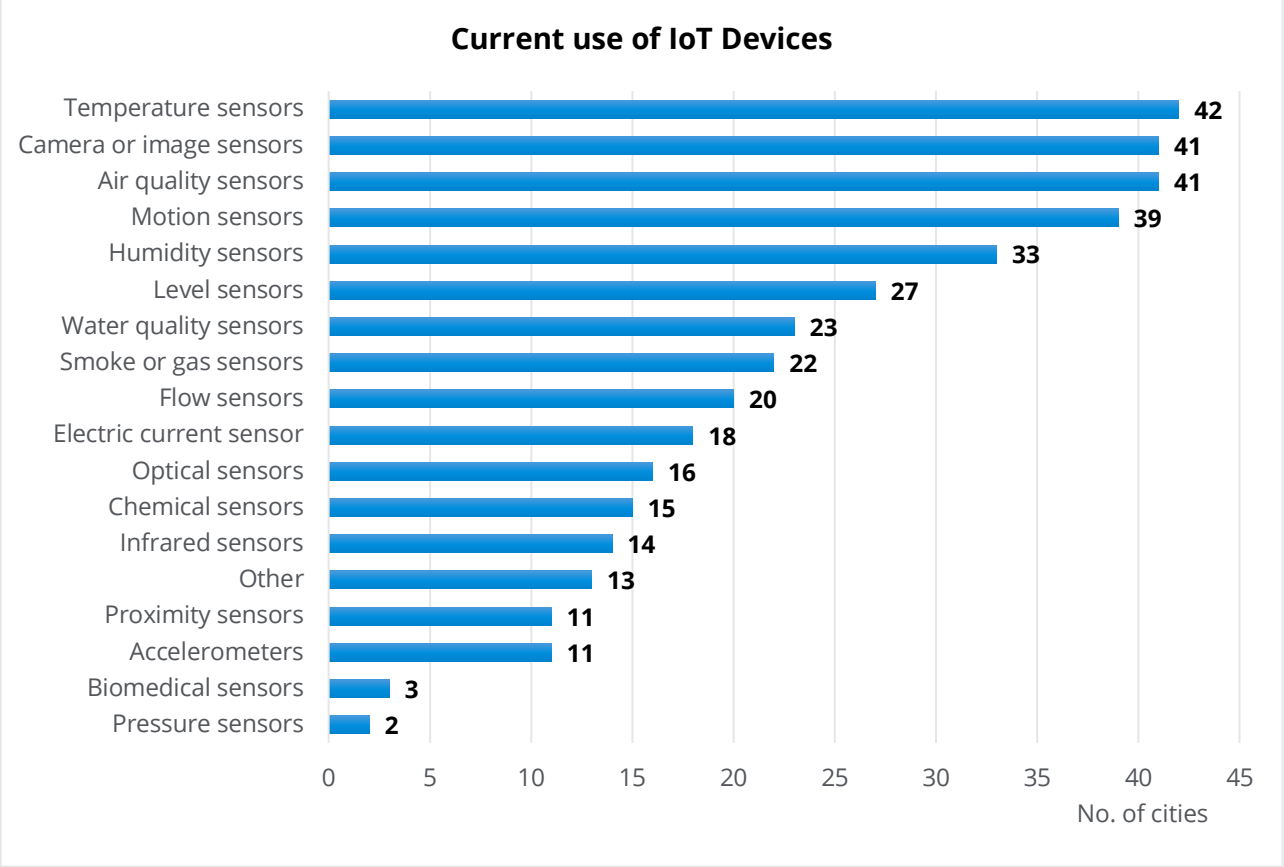
75% of the communities surveyed have 5 or more different types of IoT devices in use. For those communities, there is a wide variety in the type of devices deployed supporting their strategic priorities and use cases. The figure below illustrates the diversity of IoT devices typically used by smart cities and communities.

Figure 22 - IoT Devices



Source: Consortium (based on various experiences with smart city projects)

Figure 23 - Current use of IoT Devices



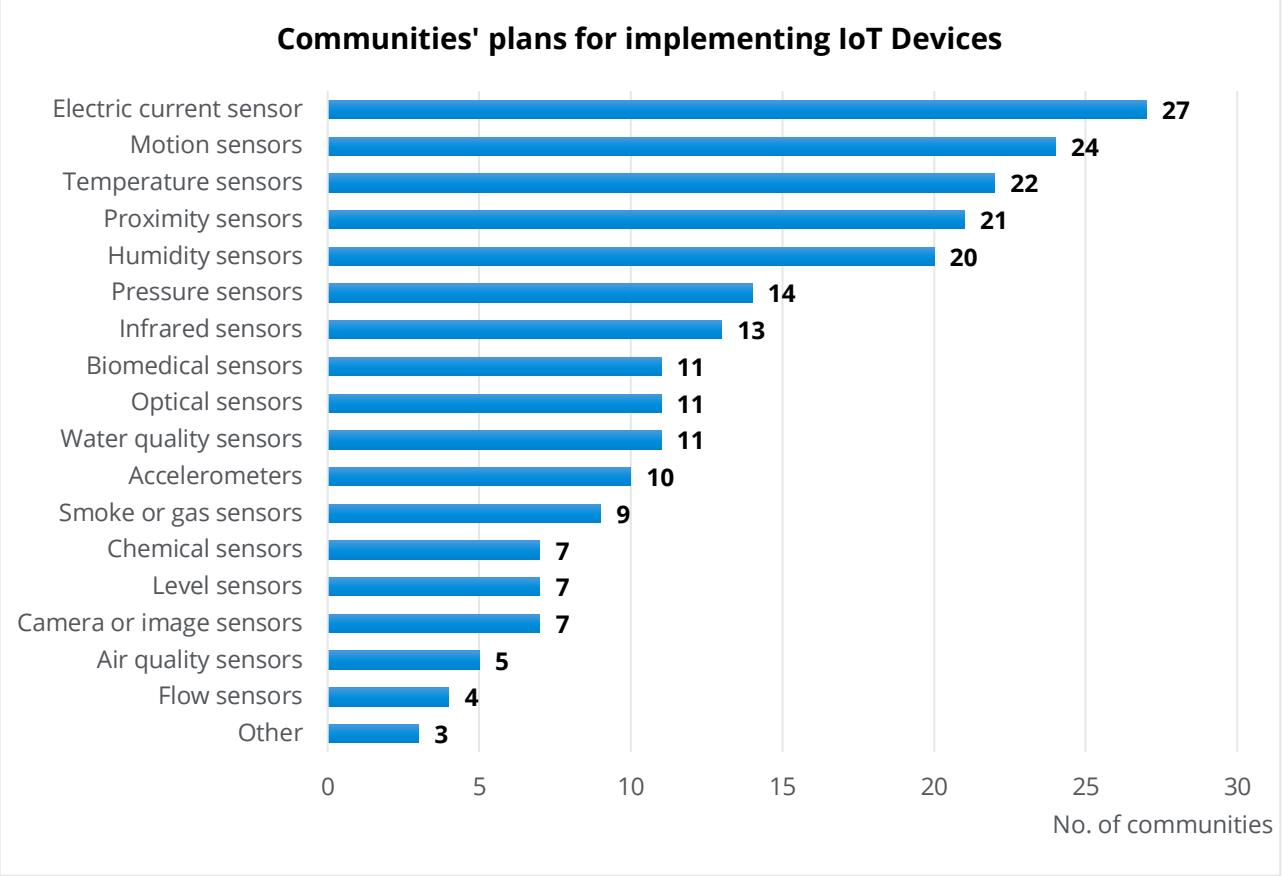
Source: Consortium (Helpdesk assessment results)

Key message #10:

75% Of the communities have **5 or more different types of IoT devices** in use.

The devices that have seen the **most use are temperature sensors, air quality sensors, cameras and motion sensors.** This points to the use of these devices for either urban planning, environmental purposes (e.g., air quality and temperature monitoring) or for traffic control. Additionally, **78% of communities have ongoing plans to implement IoT devices within the coming years**, with major efforts aimed at electric current, motion, temperature, proximity and humidity sensors.

Figure 24 – Communities' plans for implementing IoT Devices



Source: Consortium (Helpdesk assessment results)

Despite some considerable use of IoT Devices, there is room for improvement in connected IoT deployment. In particular, only 43% of communities have a platform to manage their devices in an integrated manner, and only 23% of communities have a strategy to plan for the evolution of device management over time. This is also resembled in the overall maturity score of 3.1 out of 5 for IoT Devices.

Key message #11:

78% Of the communities have **ongoing plans to implement IoT devices** within the coming years

1.7.2. Storage, Compute, and Network

The digital development of a city and the implementation of a digital twin rely heavily on **robust compute, storage, and network capabilities**. These elements form the backbone of the infrastructure required to manage the vast amounts of data generated from various sources within a city. Effective data transmission, storage, and utilisation are crucial for real-time monitoring, analysis, and decision-making processes that drive smart city initiatives.

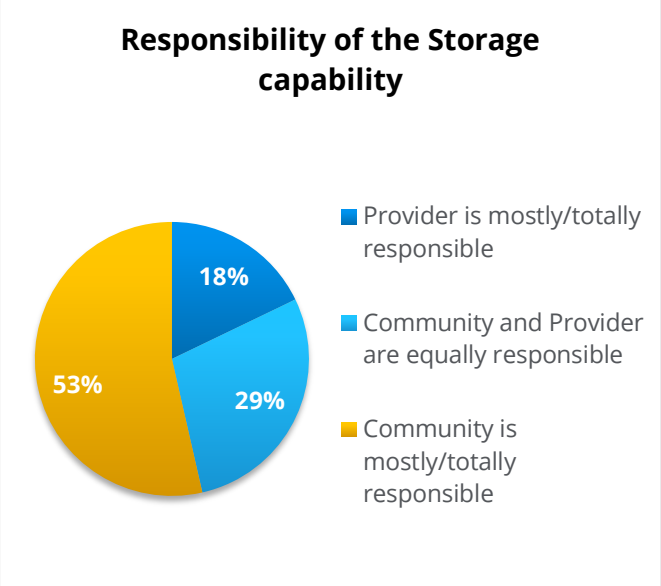
Compute capabilities enable the processing of large datasets and the execution of complex algorithms necessary for predictive analytics and simulations. **Storage solutions** are essential for retaining historical data, which is vital for trend analysis and long-term planning. **Network capabilities** ensure seamless data flow between different systems and devices, facilitating real-time communication and coordination.

Together, these capabilities support the creation and maintenance of a digital twin, providing a comprehensive and dynamic model of the city that can be used to optimise operations, enhance services, and improve the quality of life for residents.

1.7.2.1. Storage

The **storage solution and capabilities** of a community can either be the responsibility of the community itself or outsourced to an external partner. Both approaches exist among communities, with most opting for a hybrid format of the two, showing their commitment to being responsible for their storage but indicating a need for external knowledge. However, there is a strong preference for keeping most responsibilities in-house, with 82% having half or more of the responsibility assigned to the community.

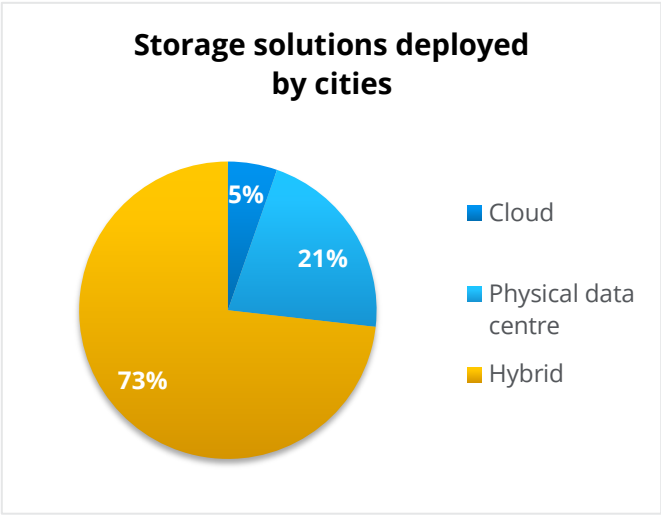
Figure 25 - Responsibility of the Storage capability



Source: Consortium (Helpdesk assessment results)

Out of 56 communities surveyed, 53 utilise on-premise storage solutions, while 44 make use of cloud storage. Notably, **73% employ a hybrid approach**, leveraging both on-premise and cloud storage to some extent. Despite the widespread adoption of cloud storage, the maturity of on-premise solutions is slightly higher, indicating a preference or reliance on traditional storage methods.

Figure 26 - Storage solutions deployed by communities



Source: Consortium (Helpdesk assessment results)

Out of the five storage-specific capabilities important for a storage solution (effectiveness, redundancy, scalability, low latency and adaptability), most communities indicate a fairly similar strength across these five skills, except for adaptability for on-premise solutions. This indicates **potential problems faced in adapting storage to changes in technology and requirements**, as opposed to cloud solutions, where these possibilities are inherited by the system used.

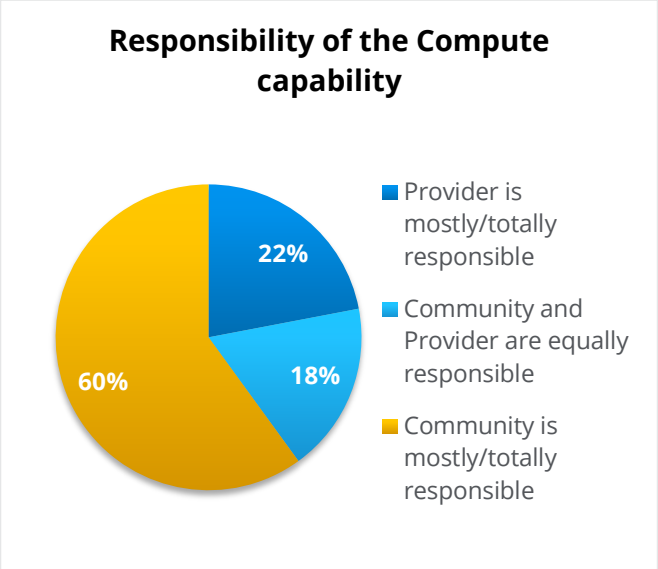
Key message #12:

73% Of the EU communities make use of a hybrid solution for their storage requirements, combining on premise and cloud storage to meet their requirements.

1.7.2.2. Compute

The division of roles for the Compute capability is similar to those for Storage, with either the community or a provider taking on the responsibilities. Conclusions also resemble those of storage, with many hybrid formats in use. However, **there is a clear preference for retaining at least half of the responsibilities within the community, accounting for 78% of the communities**.

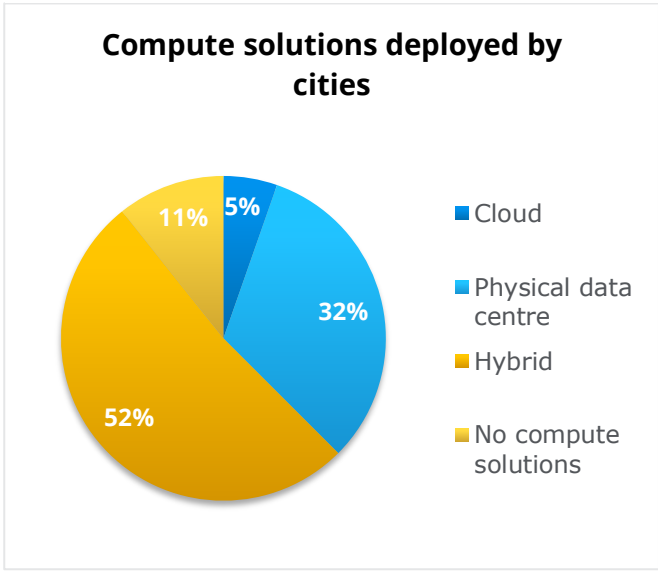
Figure 27 - Responsibility of the Compute capability



Source: Consortium (Helpdesk assessment results)

In general, the maturity level of compute solutions remains low, with an average score of 2.7. In terms of their capabilities, **32% of the communities use on-premise solutions, and 5% utilise cloud-based computing, with 52% of the communities combining the two**. Similar to storage, the maturity of on-premise compute solutions is slightly higher.

Figure 28 - Compute solutions deployed by communities



Out of the four compute-specific capabilities important for a computing solution (effectiveness, redundancy, scalability, and adaptability), most communities indicate a fairly similar strength across these five dimensions, except for adaptability for on-premise solutions. This suggests that on-premise solutions may face challenges in adjusting computing power to accommodate evolving technology and requirements, whereas cloud solutions inherently offer greater flexibility in adapting to such changes.

Communities that leverage cloud computing have access to advanced technologies such as:

- **Edge computing** (employed by 25% of the communities): distributed computing paradigm where data processing and analysis occur closer to the data source or "edge" of the network rather than in centralised data centres or cloud environments.
- **Serverless computing** (employed by 47% of the communities): a model where the cloud provider automatically manages the infrastructure required to run applications.
- **Containers** (employed by 53% of the communities): lightweight, portable units that package an application and its dependencies, allowing it to run consistently across different environments.
- **Virtual machines** (employed by 94% of the communities): fully isolated environments that run an entire operating system (OS) and simulate hardware, allowing multiple OS instances to run on a single physical machine

Finally, the communities with the lowest technical maturity level generally do not use cloud computing, highlighting a key area for potential development.

Key message #13:

52% Of the EU communities make use of a hybrid solution for their computing requirements, combining on-premise and cloud solutions to meet their needs.

1.7.2.3. Network

Network capabilities received a generally good score of 3.4. Most communities (34 out of 56) are capable of handling a large number of devices with substantial concurrent traffic exceeding 1 Gbps. Additionally, 5G technology is enabled to some extent in 32 communities, with ongoing efforts in 26 communities to enable or improve 5G infrastructure. However, communities feel least confident in advanced technologies, particularly Edge Computing and 5G readiness.

Overall, the assessment results demonstrate that while there is a foundational infrastructure in place, **there is room for improvement in the maturity and adoption of advanced technologies, particularly in storage and compute capabilities**. Enhancing these areas will be crucial for the successful digital development of communities and the implementation of effective digital twins.

1.7.3. Infrastructure security

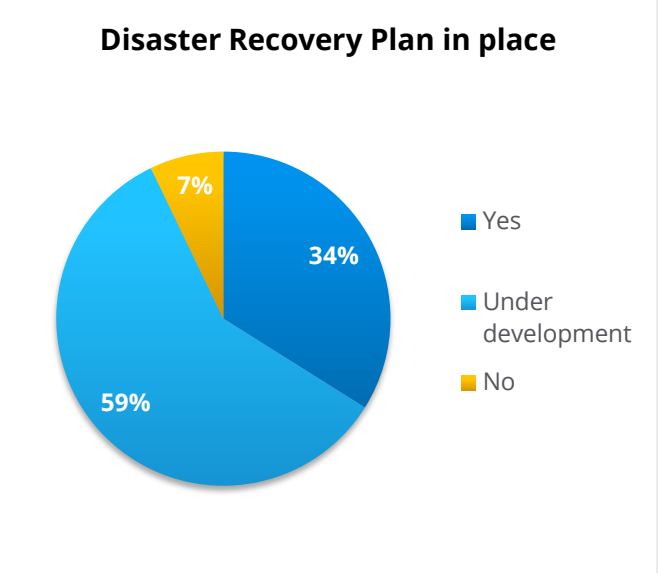
Infrastructure security is regarded as a priority topic by various communities, and EU communities' focus on this area positions them slightly above the average level obtained for technical capabilities, approaching advanced proficiency, with a score of 3.4 out of 5. All communities but one have policies and procedures in place or are developing them at this moment, further emphasising the fact that most communities recognise its value. Policies that have been implemented the most are access control policies, network security policies, endpoint security policies and data protection and privacy policies, each pointing to a distinct area of potential vulnerabilities.

Additionally, most communities (89%) have **Authentication Federation tools** in place; however, nearly half have implemented them only partially, failing to apply them across all systems. This suggests that while there is awareness of the benefits of centralised authentication and identity management, gaps in full implementation remain. Addressing these gaps would enhance security, streamline user access, and reduce vulnerabilities associated with fragmented authentication mechanisms.

In terms of proactive cyber security practices, it is observed that most communities, **around 75%, systematically conduct vulnerability assessments and security events monitoring**, which contributes to protecting IT infrastructure by identifying, assessing, and mitigating risks.

One area in which most communities were **lacking is disaster recovery**, meaning the ability to continue working in the event of a disaster, either natural or digital. **Only 34% of communities have a plan for dealing with these disruptive events**, and fewer have a business continuity plan or a backup solution in place.

Figure 29 - Disaster Recovery Plan



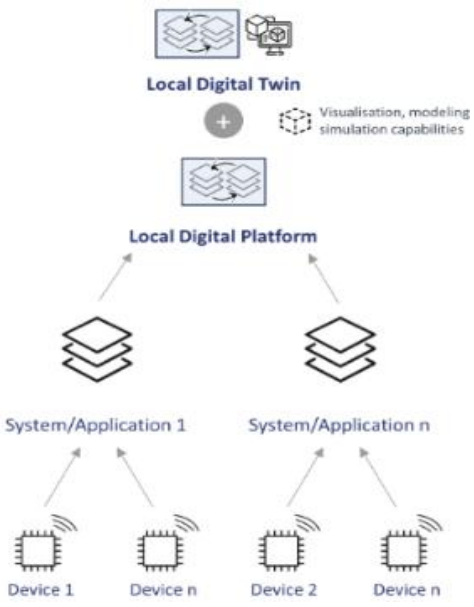
Source: Consortium (Helpdesk assessment results)

Key message #14:

98% Of the EU communities have policies and procedures in place for the cybercecurity of their infrastructure.

1.8. Local Digital Platform

Figure 30 - Understanding the concept of LDP and LDT



LDPs are the beating hearts of EU communities' digital transformation as they connect and analyse all data.

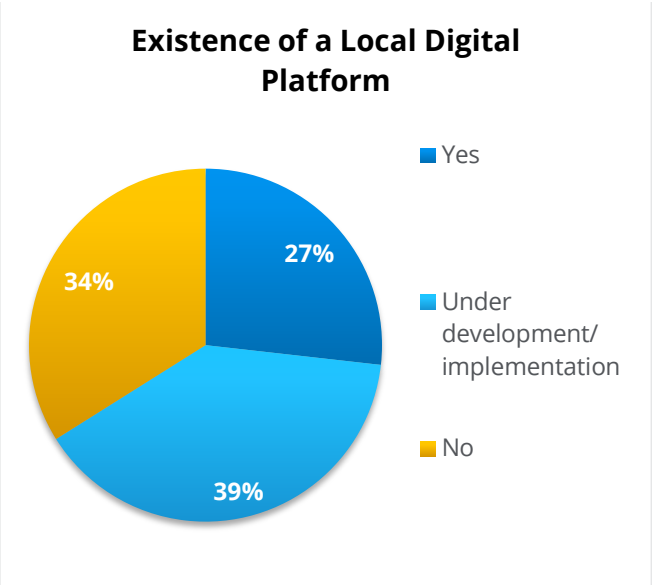
They serve as a **centralised platform** for collecting data from various **systems and applications, which also gather data from multiple IoT devices** and other types of data sources**.** LDPs play a crucial role in supporting smart community initiatives by providing a **unified and accessible data** repository for decision-making and planning community services and operations.

LDPs are also essential in the development of LDTs, as they establish the fundamental infrastructure for data collection, integration, and management, which supports visualisation, modelling, and simulation within LDTs.

LDTs serve as **virtual models** of a community's physical assets, leveraging data, analytics, and machine learning to power **simulation models** that can be dynamically **updated in real-time**, reflecting changes in the physical assets.

Source: Consortium

Figure 31 - Existence of a Local Digital Platform



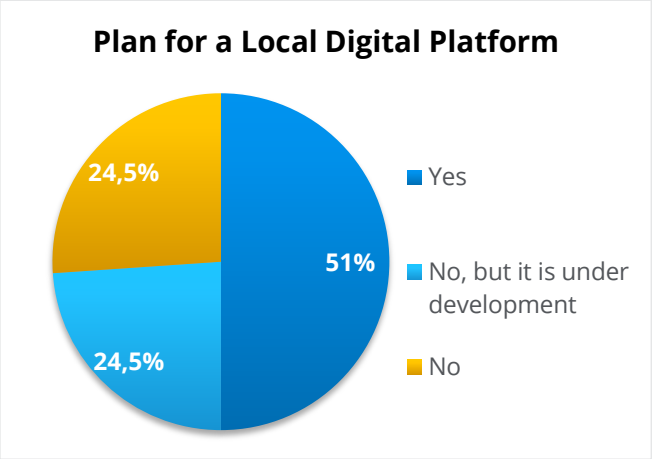
Source: Consortium (Helpdesk assessment results)

Of the 56 EU communities, **more than a third lack an LDP**, meaning they currently lack a centralised system for integrating data from various sources, such as applications, systems, and IoT devices. In contrast, **two-thirds either have an LDP or are in the process of developing one**. This highlights a significant number of EU communities adopting and advancing centralised digital systems, reflecting acknowledgement of the value smart city infrastructure brings in streamlining operations and enhancing public services.

Key message #15:

66% Of the EU communities either have a LDP or are in the process of developing one.

Figure 32 - Plan for a Local Digital Platform



Source: Consortium (Helpdesk assessment results)

Out of the communities analysed, 51% have a plan for an LDP. Interestingly, out of the 19 EU communities without an LDP, nearly half (8 communities) have no plans to develop one. This lack of planning underscores the pressing need for these communities to prioritise digital transformation strategies. Without a roadmap, they risk falling behind in leveraging technology to improve public services, enhance connectivity, and support economic development.

Key message #16:

51% Of the EU communities have a LDP plan in place or are in the process of developing one.

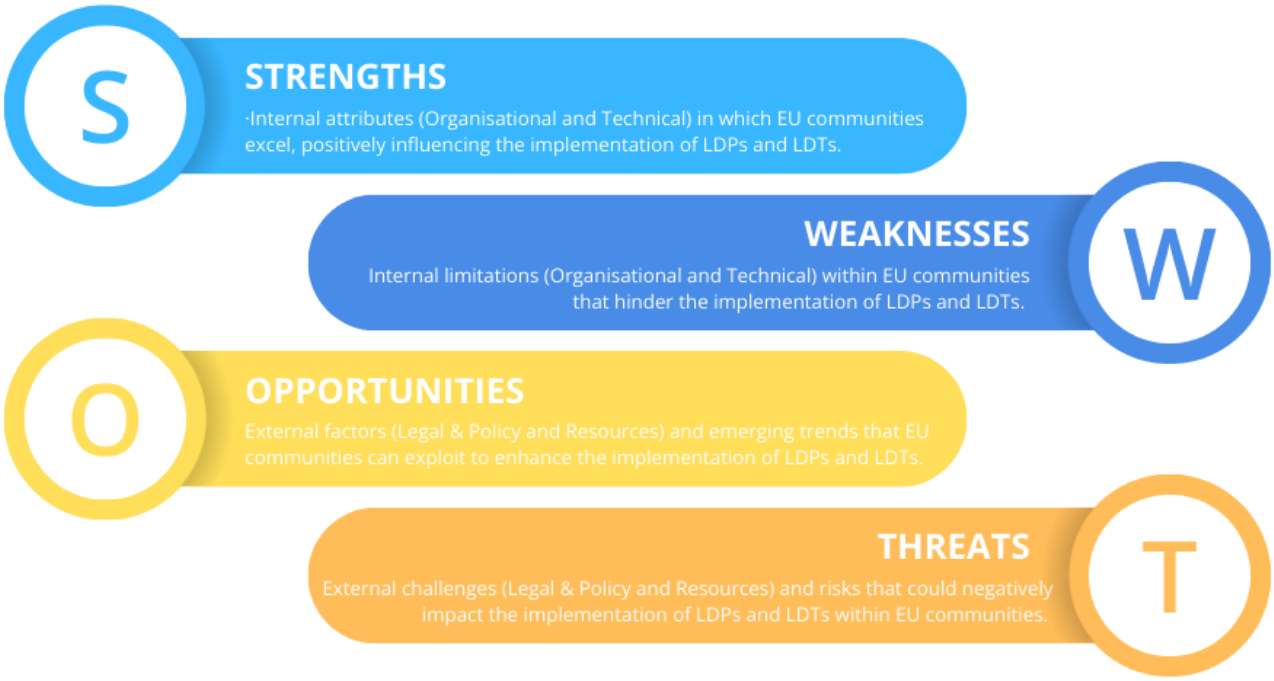
2. SWOT Analysis

2.1. Rationale guiding the SWOT analysis

This chapter provides a practical overview of the current status of the digital infrastructure of EU communities in view of future implementation (or improvements) of LDPs and LDTs to support their operations and strategic priorities. It highlights the internal and external factors that affect the implementation of LDPs and LDTs. In doing so, we summarise the strengths and weaknesses observed in terms of their organisational and core IT capabilities that are essential to prepare for LDPs and LDTs, as well as market drivers and constraints, drawing insights from the sample of 56 communities covered so far by the Online Procurement Helpdesk services.

Additionally, it will identify opportunities and threats, complementing the analysis with the wider context of legal, regulatory and policy frameworks, funding and financing landscape, and procurement, among others. This approach aims to contribute to a greater understanding of both the current landscape and future potential. It will serve as input for EU policy implications and the next steps for the EU strategy towards LDTs.

Figure 33 - Rationale for the SWOT Analysis



Source: Consortium

In the next sub-sections we delve into the specific strengths, weaknesses, opportunities and potential threats for LDT development in communities in their early stages of digital transformation.

2.2. Strengths and Weaknesses

The analysis of strengths and weaknesses is based on two key sub-criteria: *Organisational* and *Technical* aspects relevant to the development of the digital infrastructure in line with established strategic priorities

and sectors for digital transformation. While **organisational aspects** refer to the city's internal organisational structure, governance structures and mechanisms that may drive (or limit) digital transformation efforts, **technical aspects** concern the technical maturity of EU communities by evaluating IoT devices, systems, and applications that drive digital transformation while ensuring adherence to modern standards. Additionally, it assesses storage, computing, and network resources, which are crucial for efficient data management.

STRENGTHS | Organisational aspects

S

- **General adoption of digital strategies**: 86% of surveyed EU communities have either established or are currently in the process of developing a digital strategy. This widespread adoption underscores the fundamental role that digital transformation plays in shaping the strategic priorities of EU communities. The creation of a digital strategy is crucial for guiding the implementation and growth of LDPs and LDTs.
- **Increasing transparency on strategic digital priorities**: 65% of EU communities have published or plan to publish their digital strategies, promoting openness and accountability. This trend ensures that stakeholders and citizens are informed and engaged, facilitating shared learning and benchmarking among communities and opening up opportunities for new collaborations with the innovation ecosystem on these topics.
- **Governance optimisation around digital priorities:** 68% of EU communities have established governance structures to oversee digital strategies. The emerging roles of CDOs and CTOs provide critical leadership and strategic direction, ensuring alignment across digital projects.
- **Strong oversight of data management**: 61% of EU communities manage their data internally, while a limited number of communities outsource this responsibility. Internal data management ensures stronger governance, control and usability over the collected data.
- **Significant progress in data governance frameworks**: Two-thirds of EU communities have or are developing a data governance framework, highlighting the importance of structured data management. Effective data governance ensures better interoperability and informed decision-making across departments and operations.
- **Increasing collaboration through shared digital resources**: 82% of EU communities plan to share digital resources such as data, platforms, and other IT infrastructure components with partner communities (e.g., within the same region). This collaborative approach optimises the use of digital infrastructure and reduces costs, particularly benefiting smaller communities with limited resources.
- **General adoption of LDP plans:** 51% of EU communities either have an LDP plan in place or are in the process of developing one, reflecting acknowledgement that the LDP streamlines operations and enhances public services.

STRENGTHS | Technical aspects

S

- **Adoption of real-time data collection is gaining traction**: 75% of EU communities have 5 or more different types of IoT devices in use. Additionally, most EU communities have ongoing plans to implement more IoT devices within the coming years. 69% of communities rely on more than five data collection sources, which include IoT sensors, community systems or databases and regional and national government systems.

- **Scalable storage solutions**: EU communities are leveraging on hybrid storage solutions to ensure scalability and flexibility: 73% of communities have cloud-based storage solutions, allowing them to dynamically manage their data needs.
- **Progress in network infrastructure will enable digital growth**: EU communities report an average network maturity score of 3.4 out of 5, which is a critical enabler for expanding digital initiatives and smart city solutions.
- **Advancements in following GDPR**: All communities follow GDPR, and most have the relevant policies and procedures in place to do this in a structured way throughout their departments. However, a majority of communities still need to implement tools that support compliance with GDPR.
- **Proactive cyber security measures**: 98% of the EU communities have established policies and procedures to secure their infrastructure. Approximately 75% regularly conduct vulnerability assessments and monitor security events.
- **Expanding LDPs to support data-driven transformation**: Two-thirds of EU communities have either developed or are actively building an LDP, sometimes targeting specific use case(s) or already envisaging future integration of various city domains/verticals. LDPs will play a crucial role in integrating data from diverse applications, systems, and IoT devices, serving as a key enabler for the development of LDTs and other smart city innovations.

WEAKNESSES | Organisational aspects



- **Fragmentation is a blocker for unifying digital priorities**: 45% of EU communities with a digital strategy report difficulties in aligning initiatives and priorities from different departments in the field of digitalisation due to internal fragmentation. This misalignment can delay the adoption and effective execution of digital strategies.
- **Absence of Data Strategy**: Only 31% of EU communities have active data strategies, showing a significant gap in coordinated data management. This shortfall undermines efforts to optimise data use for informed decision-making and efficient service delivery.
- **Lag between the announcement of new digital-focused governance structure(s) and their actual integration in the governance of the community**: Multidisciplinary teams focused on digital and smart city strategies are taking longer to become formalised, with 54% of teams not yet formally integrated into the governance structure. This delay presents an obstacle to the effective coordination and implementation of digital initiatives, impacting resource allocation and budgeting within municipalities.
- **Lacking monitoring mechanisms and defined KPIs**: Only 41% of EU communities are currently working on defining KPIs, and 55% of those still lack dedicated monitoring and reporting mechanisms. This gap in tracking hinders the ability to measure progress and ensure accountability in digital transformation initiatives.
- **Lacking a disaster recovery plan**: Only 34% of communities have a plan for dealing with disaster recovery events, and fewer have a business continuity plan or a backup solution in place.
- **LDTs are still absent from the majority of strategic plans**: 65% of EU communities do not incorporate LDTs in their strategic planning yet, revealing a there is still a gap in the adoption of comprehensive digital solutions for urban management.
- **Budget constraints**: Smaller EU communities face budget constraints that hinder the establishment of multidisciplinary teams focused on digital priorities, as well as the development of necessary skills for their profiles.

- **Lack of capacity to adopt procurement practices in some communities**: Even when funding is identified, responding to calls requires dedicated resources, which are often limited.
- **Organisational limitations in accessing and utilising EU funding for procurement**: Communities often struggle to identify suitable EU funding sources for LDT procurement due to the complexity and fragmentation of available programs. While multiple funding options exist, locating the right opportunity and understanding eligibility criteria can be difficult.

WEAKNESSES | Technical aspects

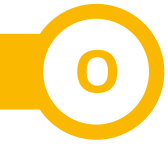


- **Lack of advanced IoT device management strategies**: While 75% have 5 or more different types of IoT devices in use, only 20 communities have a platform for device management, and just 10 communities have a strategy for managing them, lacking operational maturity. Only 17 communities have a sensor registry.
- **Data analysis capabilities remain underdeveloped:** While data collection and retention are relatively mature, data analysis lags behind, limiting communities' ability to extract valuable insights for decision-making. Only 56% of communities use data analysis tools.
- **Interoperability shows growing awareness but limited adoption**: Although many communities acknowledge the importance of interoperability and are working on developing relevant measures, it has not yet been fully standardised or integrated across all communities. The interoperability maturity score for communities is 36.7, indicating that they are still in the early stages of digital transition.
- **Limited maturity in Computational capabilities**: The average maturity score for compute solutions is only 2.7 out of 5, with many communities struggling with scalability and adaptability, particularly for on-premise solutions.
- **Integration issues with legacy systems**: Implementing newly developed or procured digital solutions often requires integrating them with existing large IT systems and infrastructure, which can pose interoperability barriers and integration challenges.
- **Technical procurement knowledge gap:** EU communities report challenges in identifying and translating their procurement needs into detailed technical specifications. This is primarily linked to technical expertise gaps, which could lead to suboptimal outcomes in the procurement processes.
- **Difficulties in proposal evaluation and provider selection**: EU communities often struggle with establishing clear criteria for evaluating proposals and selecting providers, making procurement decisions and provider selection challenging.

2.3. Opportunities and Threats

The analysis of opportunities and threats is based on Legal & Policy and Resource aspects. In this subsection, we will therefore examine the legal, regulatory and policy guidance that impacts (negatively or positively) the digital transformation of EU communities and, more particularly, the development of LDTs. We explore available funding and non-funding options to assess potential opportunities or risks that could influence the development of LDTs.

OPPORTUNITIES | Legal, Regulatory & Policy aspects



- **EU legal and regulatory frameworks promote a safer and more efficient use of AI and other technologies key for simulation capabilities of digital twins**: EU legislation on data protection (e.g., GDPR, Data Act, Data Governance Act, Fair AI (e.g., AI Act), the European Interoperability Framework for Smart Cities and Communities (EIF4SCC) and the Interoperable Europe Act provide important guidance to governments, and enable a more efficient, interoperable and secure environment for integrating data from various data sources, and for developing LDTs that centralise this information.
- **Collaboration opportunities enhance experimentation, innovation and knowledge sharing among communities**: Participation in the Living-in.EU initiative connects stakeholders and keeps them informed about the latest developments in data spaces, interoperability, and LDTs, encouraging collaboration and advancing the LDT ecosystem. Other initiatives include the Data Space for Smart and Sustainable Cities and Communities (DS4SSCC), the EU Mission on 100 Climate-neutral and Smart Cities (NETZERO CITIES), Urban Innovative Actions (UIA), and the Intelligent Cities Challenge (ICC).
- **Growing focus on standardisation and compliance in procurement:** The increasing emphasis on open standards, interoperability frameworks, and EU regulations in digital procurement is enhancing transparency and competition. This reduces dependency on single vendors, creating a more dynamic and resilient LDT market. In addition, synergies with EU initiatives like the Big Buyers Programme within the CoP Sustainability in ICT (Digital), Smart Cities Marketplace, the European Sustainable Procurement Network (Procura+), EIC Innovation procurement programme with the ongoing projects SPIN4EIC and INNOMATCH focused on Innovation procurement to engage startups and SME´s as top-notch innovators further support procurement efforts across communities, fostering promoting procurement that can implement digitalisation.
- **Joint procurement as a driver for LDT adoption for smaller communities:** By pooling resources and coordinating purchases among contracting authorities, municipalities can reduce financial and administrative burdens, gain better contract terms, and ensure compliance with EU procurement regulations. This procurement method could potentially foster also greater standardisation, interoperability, and scalability of LDT solutions, making them more accessible to a wider range of cities and regions. Additionally, joint procurement can stimulate market demand, encouraging suppliers to develop more tailored and competitive solutions, ultimately driving innovation and accelerating the digital transformation of European communities.
- **Certification schemes will potentially ensure compliance and foster market development**: The development of a MIMs Plus Conformance and Certification Scheme, to be managed by the European Digital Infrastructure Consortium (EDIC) or another EU entity, will allow communities to ensure that smart city solutions comply with MIMs Plus standards, promoting the growth of a thriving LDP marketplace driven by certified solutions.
- **Frameworks for assessing LDT maturity support an evolutionary approach**: Initiatives such as the Horizon 2020-funded DUET Maturity Model help cities and regions evaluate their digital twin capabilities and set gradual development goals.
- **Digital maturity indicators enable continuous performance tracking**: The existence of self-assessment tools like LORDIMAS and DIGISER allows EU communities to monitor and benchmark their digital maturity over time, facilitating critical assessments on the progress over time and the required improvements supporting LDT adoption.

OPPORTUNITIES | Resource aspects



- **Multiple EU funding mechanisms support digital transformation**: Programs such as Horizon Europe, DEP, Connecting Europe Facility (CEF) building blocks, European Social Fund Plus (ESF+), and

Cohesion Funds offer financial support for LDT-related projects, ensuring resources for innovation, infrastructure development, and research.

- **Support beyond direct funding**: Instead of providing direct financial grants, the EU offers a variety of support services, tools, and technical assistance to help communities develop LDTs and LDPs. This approach democratises access to resources while encouraging communities to strategically assess their needs and selectively apply for funding. As a result, communities can better prepare for funding opportunities, ensuring a more efficient and targeted use of financial support.
- **Active initiatives are designed based on continuous governance and focused on AI-powered and user-centric approaches**: As part of the Digital Decade Policy Programme 2030, the European Digital Infrastructure Consortium (EDIC) on the CitiVERSE intends to "build an EU ecosystem of advanced AI solutions for cities" and already counts with 14 EU Member States. Besides the Online Procurement Helpdesk, there will be support services provided through an EU LDT Toolbox composed of various open-source tools and AI models for communities to freely use and integrate into the contexts of their urban challenges.
- **Efforts to enhance the digital skills of public sector employees**: Several initiatives and organisations are dedicated to enhancing the digital skills of public sector employees, enabling them to navigate the digital transformation of their public services. Notable examples include the European Institute of Public Administration (EIPA), the European Public Administration Network (EUPAN), the European Digital Innovation Hubs (EDIHs), and the EU eGovernment Network.
- **Leveraging successful use cases and best practices**: The growing network of EU communities implementing LDTs and LDPs enables knowledge sharing, helping new adopters learn from established projects, avoid pitfalls, and accelerate their digital transformation.

THREATS | Legal, Regulatory & Policy aspects

T

- **Challenges to cope with standards evolution and regulatory changes**: Future revisions to AI regulations, data governance laws, interoperability regulations or sector-specific policies could impose additional compliance burdens and a learning curve to communities, requiring costly adjustments to existing LDT infrastructures.
- **Diverging interpretations in national regulations and delays**: While the EU aims for harmonised digital policies, differences in how member states implement regulations (e.g., GDPR enforcement, AI Act interpretation) can create legal uncertainty for LDT initiatives operating across borders.
- **Liability, accountability and trust concerns**: Ambiguities in legal responsibility for AI-driven decisions, digital twin simulations, and predictive analytics could discourage public authorities and private actors from adopting LDT solutions.
- **Risk of vendor lock-in in procurement:** Limited contractualisation conditions, standardised strategy, and certain types of public procurement procedures with technological providers increase the risk of vendor lock-in, where communities become dependent on a single supplier due to proprietary technologies or restrictive procurement contracts, reducing flexibility and hindering innovation.

THREATS | Resource aspects

T

- **Dependency on EU funding cycles**: Many LDT-related initiatives rely on EU grants or tenders (e.g., Horizon Europe and DEP). A shift in funding priorities or reduced financial support could disrupt ongoing projects. Additionally, varying funding and financing mechanisms adopted by EU member

states and communities for LDPs and LDTs contribute to disparities in digital development across the EU.

- **Complexity and fragmentation of available programs**: This leads (or may lead) to challenges in accessing and utilising EU funding for procurement.
- **Limited experience with EU interoperable digital solutions**: Many EU communities have low adoption rates of EU-backed digital tools and frameworks (e.g., ISA², CEF Digital Building Blocks), leading to inefficiencies in integration and standardisation efforts[5].
- **Slow uptake of data-sharing initiatives and digital ecosystems**: While European Data Spaces and interoperability initiatives exist, many communities struggle to participate due to technical, legal, or operational barriers, limiting the potential for large-scale collaboration.

[5] Source: European Commission: Directorate-General for Communications Networks, Content and Technology, Robalo Correia, A., Sousa, M., Mulquin, M., Santos, F. et al., *Mapping EU-based LDT providers and users*, European Commission, 2023, **https://data.europa.eu/doi/10.2759/547098**

3. Policy implications and next steps for the EU Strategy towards Local Digital Twins

3.1. 10 Recommendations on the way forward

Based on the key conclusions highlighted in the SWOT analysis, this section puts forward ten cross-dimensional recommendations to guide the EU strategy towards LDTs, focusing on aspects that relate to funding, knowledge-sharing, capacity-building, experimentation and alignment with national and regional levels. These recommendations could contribute to scalability and EU-wide replication.

Recommendation #1: Continue to leverage the Online Procurement Helpdesk service to centralise information and support on the procurement of smart city solutions and technologies

The EC plays a crucial role in supporting EU communities with procurement guidelines, templates, and other resources, namely through the Online Procurement Helpdesk for Smart Communities. It is essential to regularly update these procurement templates and guidelines to align with any regulatory updates and the latest MIMs Plus developments.

The EC could also follow up on participating EU communities and their LDT development and incentivise them to make use of the available procurement documents and/or of the upcoming Local Digital Twin Toolbox open-source tools and resources that will become available to all communities. Additionally, leveraging the developed infrastructure and support model, the EU can continue to provide ad hoc procurement support and assistance through the Online Procurement Helpdesk.

Recommendation #2: Consider providing funding to support the deployment and implementation phases of the communities serviced by the Online Procurement Helpdesk

Communities have acknowledged the importance of a LDP and LDT. However, especially for smaller and less digitally mature city administrations, the level of resources (capacity and financial) is often considered insufficient to invest and implement the digital infrastructure components they need to make that evolution. In the same way, they often lack resources to ensure gradual upgrades and maintenance of LDP and LDT.

Recommendation #3: Provide learning and training resources, toolkits and/or supporting consultancy services to guide effective data management and governance within cities and communities

As a large portion of communities are themselves responsible for data management, activities are required to improve data capabilities. Toolkits and best practices from frontrunning communities, particularly in the area of data governance, need to be further developed. These insights should be transferred to cities that are lagging, helping them bridge the gap.

The focus should be on helping communities progress from low to basic capabilities in critical areas such as data analysis, privacy-enhancing technologies, GDPR compliance, and interoperability (including the Interoperable Europe Act). Additionally, improvements in data retention and collection practices should be prioritised to move communities from average to more advanced utilisation levels.

Recommendation #4: **Communicate and prioritise the LDT concept in other EU projects and initiatives**

LDTs can be powerful tools across various types of use cases in cities and communities, contributing to EU Green Deal and Digital Decade objectives. For this reason, their importance and role could be communicated more prominently in the context of projects and initiatives such as the EU Mission on Climate-neutral and Smart Cities, the Intelligent Cities Challenge, Interoperable Europe, and Urban Innovative Actions, among others. Living-in.eu and the EDIC on the CitiVERSE will certainly also boost the importance of LDT development.

Recommendation #5: **Foster the coordination and monitoring of digital infrastructure topics such as LDT, aligning EU and national priorities and investments**

EU-national alignment through the EDIC, Smart Communities network and any other existing governance mechanisms/structures is crucial to maximise synergies and impact from funding and other collaboration initiatives. There are positive developments in some EU Member States in terms of bringing smart cities and LDT to the national agenda linked to digital transformation[6]. Making use of various funding instruments and programmes in an integrated and complementary way (DEP, CEF, Structural funds, Recovery and Resilience Plans) could lead to efficiency gains.

Recommendation #6: **Incentivise the use of EU digital maturity tools, namely LORDIMAS, at the national level for a harmonised understanding of digital maturity in communities across the EU**

EU Member States, with the support of the EC and the Committee of the Regions (CoR), should promote the use of LORDIMAS in the assessment of their "as-is" situation when it comes to new city digital strategies and action plans on this topic. This enables a consistent analysis of the gaps of EU communities in terms of organisational and technical aspects.

Recommendation #7: **Promote learning sessions between EU communities and the innovation ecosystem around more sophisticated simulation and visualisation capabilities of digital twins by type of use case**

As EU communities advance in their roadmap towards LDP and LDT, it is important to generate awareness around the use of algorithms, models and what-if scenarios to address more specific needs and use cases they have. For example, this could concern flood-risk management, crowd management, disaster risk management, and urban planning, among others. Given the complexity and evolving nature of these topics, expertise from large companies, startups, and academia is essential. Their help is needed need to develop these models, comprehend the cloud-based requirements for real-time data integration, and establish an operational command centre to support a comprehensive view of city operations.

While 3D model interfaces are already receiving some attention from cities, the actual potential of Augmented Reality (AR)/Virtual Reality (VR) use remains unclear to most. For this reason, it is important to once again mobilise key actors in the innovation ecosystem working in this space to provide concrete use cases of how these tools can support urban planning decisions, improve civic participation, promote more immersive experiences to visitors, and lead to new experiences in schools and healthcare services of the city, just to name a few.

Recommendation #8: **Continue the promotion of Wi-Fi and 5G networks across EU communities through EU funding programmes to ensure real-time data integration and insights**

[6] You can consult some of these examples at https://op.europa.eu/en/publication-detail/-/publication/7892c228-359f-11ee-800c-01aa75ed71a1/language-en

Initiatives like WiFi4EU have proven particularly helpful, especially for smaller EU communities. Wireless (Wi-Fi, 5G, LTE) communication technologies allow devices to connect to the internet or other networks without the need for physical cables. For this reason, continued support from EU and national funding programmes is essential to minimise disruptions in the real-time transmission of data from various types of data sources connected to the LDP or LDT.

Recommendation #9: Promote learning sessions with experts on the topics of network and security to enable seamless connectivity and safe real-time data insights

As the LDP and the LDT evolve and integrate more and more real-time data from devices, it will be important to enable edge computing and 5G readiness in cities. At the same time, privacy-enhancing tools will allow for the control of the effectiveness of GDPR policies. The EC could promote workshops for cities to clarify any questions on this topic (cost, benefits, legal implications) with the participation of technology companies and legal experts on some of these issues.

Recommendation #10: Ensure the uptake of the future certification scheme for MIMs Plus-compliant solutions to create a true market in the EU based on interoperable solutions that reduce vendor lock-in and legacy system issues

The possibility of testing the compliance of vendors' technologies against a set of MIMs Plus (and/or other open standard solutions) and getting certified for it can reduce the risk for EU cities and communities of finding themselves in a vendor lock-in situation. At the same time, the incentives for vendors are that the solutions they offer could rank higher in the award criteria of tenders, for example.

4. Annexes

4.1. Methodology for Digital and Technical Maturity

4.1.1. EU Communities sample

Our sample includes 56 communities: 54 are cities, with participation registered on behalf of the municipality (or related agency), and 2 are other typologies of smart territories such as regional or metropolitan area authorities.

On a methodological note, it is also important to highlight that although our sample size is 56 communities, some strategy and technology infrastructure-related questions may not be applicable to all of them. This is due to refinements made to the questionnaires in each phase, as well as the interdependence of certain questions. In such cases, our analysis focuses on the relevant subset of the sample, excluding the "Not applicable" responses. The same rationale applies to one city that has not yet provided LORDIMAS data - their responses were considered for strategy and IT-related topics but were excluded from the overall maturity scores due to missing data.

The survey encompassed a diverse range of communities across the European Union, representing various phases of digital transformation and different sizes of public administrations. The sample included a total of 56 communities from 20 EU Member States, namely Spain, the Netherlands, Belgium, Bulgaria, Slovenia, Italy, Ireland, Hungary, Germany, Portugal, Finland, Greece, Czechia, Latvia, France, Sweden, Croatia, Lithuania, Estonia, and Belgium. Table 1 displays the breakdown of communities per country.

Table 1 - Geographical representation of communities

Country	No. of communities	% of the sample
Spain	9	16%
Belgium	9	16%
Netherlands	5	9%
Italy	4	7%
Greece	4	7%
Latvia	3	5%
Slovenia	3	5%
France	3	5%
Portugal	2	4%
Finland	2	4%
Ireland	2	4%
Hungary	2	4%
Sweden	1	2%
Slovenia	1	2%
Bulgaria	1	2%
Lithuania	1	2%
Croatia	1	2%
Czechia	1	2%
Germany	1	2%

Estonia	1	2%

Source: Consortium

The surveyed communities varied significantly in size, ranging from smaller municipalities with populations between 5,000 and 50,000 to larger municipalities and metropolitan regions with populations exceeding 500,000. This diversity ensures a comprehensive representation of different city sizes and types of public administrations, from municipalities to metropolitan regions. However, most often, the communities found themselves in between 50 thousand and 250 thousand inhabitants. The complete breakdown can be found in Table 2:

Table 2 - Size of included communities

Population size	No. of communities	% of the sample
>5,000 <25,000	3	5%
>25,000 <50,000	7	13%
>50,000 <100,000	14	25%
>100,000 <250,000	24	43%
>250,000 <500,000	6	11%
>500,000 <1,000,000	2	4%

Source: LORDIMAS Dashboard LORDIMAS_EN

In addition to the information gathered through the assessments, this report also includes findings from workshops held with participant communities in the Helpdesk Service. These workshops focused on procurement aspects; therefore, the insights related to procurement in the SWOT analysis and recommendations reflect these findings.

4.1.2. Overall digital maturity methodology

The methodology process for assessing the digital maturity of EU communities involves the digital maturity classification of capabilities into one of six predefined maturity levels based on a matrix aligning capability scores with maturity stages.

Table 3 - LORDIMAS score classification matrix

Score Range	Maturity Level	Description
0 – 9	Nascent	Lacking digital transition
10 – 39	Reactive	Very low levels of digital transition
40 – 49	Emerging	Low levels of digital transition
50 – 69	Purposeful	Moderate levels of digital transition
70 – 89	Optimised	High levels of digital transition
90 – 100	Native	Very high digital transition

Source: LORDIMAS Dashboard LORDIMAS_EN

Methodological remark: One city in the sample did not submit LORDIMAS results and was therefore excluded from the overall digital maturity analysis.

4.1.3. Technical maturity methodology

Based on the assessment outcomes and the scores obtained, EU communities will be categorised into one of the five technical maturity levels.

Table 4 - Technical maturity levels

Score	Maturity Level	Description
1	Absent	Poor technological capabilities. However, there is willingness and/or political commitment to start its digital transformation journey, including the adoption of smart solutions.
2	Embryonic	Inadequate technological capabilities to effectively fulfil its willingness and/or political commitment to embark on a digital transformation journey.
3	Foundation	Basic technological capabilities. All capabilities need to have started their development or exploration, and Integration/ Interoperability and Data should already be in use within the community.
4	Seeker	Digital strategy, IoT devices or services and advanced technological capabilities in place, possibly already with an LDP in place or at development.
5	Growth	Implementation of an LDP already displays more mature technological capabilities, but it needs to continuously evolve towards an LDT.

Source: Consortium

Maturity levels are derived from the combined assessment of capabilities through five capability scores. For each of the seven capabilities evaluated by a scale, there is a more detailed matrix where the minimum conditions for each sub-capability are further specified.

Table 5 - Technical capability and sub-capability scoring

Score	Utilisation Level	Description
1	Not At All	Little or no understanding/ proficiency in the given capability. It's also not been explored strategically within the community yet.
2	Initial Exploration	Basic awareness or initial exploration of the capability. Communities may have started to experiment with the basic features or concepts, but they are far from having an outstanding utilisation of the capability.
3	Basic Utilisation	Functional understanding of the capability and already uses it for fundamental tasks.
4	Advanced Utilisation	Deep understanding of the complexities of the capability and can leverage its features for more complex tasks.
5	Outstanding Utilisation	Exceptional mastery and expertise of the capability. Communities at this level not only show a comprehensive understanding of the capability but can also innovate, optimise, and excel in utilising it.

Source: Consortium

Methodological remark: following the same rationale as the overall digital maturity analysis, one city in the sample did not submit LORDIMAS results and was therefore excluded from the overall technical maturity analysis.

4.1.4. Digital Strategy Maturity Scoring

The Digital Strategy Maturity Score is the combination of "Digital Strategy", which refers to the question "Does your community have a digital strategy (or equivalent) in place?" and "Strategic Vision", which refers to "How extensively defined is your strategic vision for digitalisation?".

Table 6 - Digital Strategy maturity levels

Digital strategy
Yes
Under development
Under revision
No

Source: Consortium

Table 7 - Strategy Vision maturity levels

Strategy Vision
Integrated
Dispersed
No

Source: Consortium

Table 8 - Digital Strategy Maturity Scoring

	Digital Strategy Maturity			
Capability Maturity Level Minimum Requirements	**No**	**Under Development**	**Under Revision**	**Yes**
1 Not at all				
2 Initial Exploration	✓ Digital Strategy = No	✓ Digital Strategy = Under development ✓ Strategic vision = No	✓ Digital Strategy = Under revision ✓ Strategic vision = No	
3 Basic Utilisation		✓ Digital Strategy = Under development ✓ Strategic vision = Dispersed	✓ Digital Strategy = Under revision ✓ Strategic vision = Dispersed	✓ Digital Strategy = Yes ✓ Strategic vision = No
4 Advanced Utilisation		✓ Digital Strategy = Under development ✓ Strategic vision = Integrated	✓ Digital Strategy = Under revision ✓ Strategic vision = Integrated	✓ Digital Strategy = Yes ✓ Strategic vision = Dispersed
5 Outstanding Utilisation				✓ Digital Strategy = Yes ✓ Strategic vision = Integrated

Source: Consortium



Publications Office
of the European Union

ISBN 978-92-68-33668-7 doi: 10.2759/7340057 KK-01-25-147-EN-N